



Global Connect Ethiopia – International Professional Event Hub

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DECLARATION

We are students of Adama Science and Technology University in the School of Electrical Engineering and Computing in the Department of Computer Science and Engineering. The information found in this project is our original work. And all sources of materials that will be used for the project work will be fully acknowledged.

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ACRONYM

ASTU : Adama Science and Technology University

GC : Global Connect

UI : User Interface

UX : User Experience

API : Application Programming Interface

DB : Database

DBMS : Database Management System

IPDU : Integrated Promotional Design Utility

UML : Unified Modeling Language

SDLC : Software Development Life Cycle

SRS : Software Requirements Specification

QA : Quality Assurance

CI/CD : Continuous Integration / Continuous Deployment

IDE : Integrated Development Environment

JWT : JSON Web Token

HTML : HyperText Markup Language

CSS : Cascading Style Sheets

JS : JavaScript

HTTP : Hypertext Transfer Protocol

HTTPS : Hypertext Transfer Protocol Secure

TCP/IP : Transmission Control Protocol / Internet Protocol

PWA : Progressive Web Application

CRUD : Create, Read, Update, Delete

AI : Artificial Intelligence

UXD : User Experience Design

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ABSTRACT

Global Connect Ethiopia is an intelligent web-based event management platform designed to revolutionize how professional events are planned and executed in Ethiopia. Unlike traditional systems that rely on external event organizers, the platform empowers clients to independently design, manage, and host their own events through a unified digital ecosystem. By integrating all essential components vendors, venues, professionals, travel, lodging, and communication. The system ensures efficient coordination, transparency, and real-time control from concept development to post-event analytics.

Built using modern web technologies (React, FastAPI, and MongoDB), Global Connect Ethiopia introduces innovative features such as AI-assisted event roadmaps, cost estimation tools, and chatbot-based planning support. Users can access vendor ratings, select services based on event type, and receive data-driven insights to optimize their decisions. The platform's modular design also supports multiple event domains technology summits, investment forums, research conferences, and cultural expos while ensuring enterprise-grade security for high-profile guests and government events.

By addressing the limitations of imported systems like Flawless Events and Eventbrite, Global Connect Ethiopia delivers a localized, cost-effective, and scalable solution aligned with Ethiopia's digital transformation goals. It promotes local innovation, supports small and medium enterprises, and strengthens Ethiopia's position as a regional hub for professional conferences and international collaborations.

CHAPTER 1: INTRODUCTION

1.1 Introduction

The event-management sector in Ethiopia is expanding rapidly due to economic growth, foreign investment, and increased digital adoption. Yet, professional events, technology summits, investment forums, research conferences still rely heavily on manual coordination and fragmented service networks.

Global Connect Ethiopia proposes an end-to-end digital solution that consolidates all stakeholders into a single ecosystem. By integrating event scheduling, vendor management, participant networking, and analytics, the system streamlines operations while improving transparency and accessibility.

Digital platforms such as Eventbrite or Hopin have transformed global event management, but they lack localization for the Ethiopian market, language preferences, and regional service integration. Developing a home-grown platform aligns with Ethiopia's national digital-transformation agenda and provides a learning opportunity in modern software architecture.

1.2 Background of the Project

The event industry in Ethiopia has grown rapidly over the past decade, driven by increasing interest in concerts, conferences, cultural shows, exhibitions, and international programs. However, the digital infrastructure that supports event planning and management has not kept pace with this growth. Existing event management platforms in Ethiopia are limited in scope, often expensive, and primarily focused on ticket sales rather than empowering clients to independently organize and manage their events.

Currently available systems do not provide a client-led event management approach, where users can plan, coordinate, and execute events under their own supervision. Most platforms are designed for private event organizer companies, leaving individuals, institutions, or international organizers without a centralized digital solution to oversee the full event lifecycle.

Additionally, the lack of coordinated event information frequently leads to scheduling conflicts, where two or more major events such as concerts or cultural nights occur simultaneously. This results in audience distribution issues, financial losses, and attendee dissatisfaction. For example, large-scale events by artists or international institutions may overlap due to the absence of a unified event visibility system.

Furthermore, essential processes such as licensing, security clearance, permit approval, and official communication with government entities are still largely manual. International hosts, private companies, and regional organizers face bureaucratic delays because there is no integrated digital platform to submit proposals, obtain permissions, or manage compliance requirements.

These gaps highlight the need for a unified, user-driven, technology-enabled event management platform that streamlines planning, improves transparency, and enhances coordination among stakeholders.

1.3 Statement of the Problem

The current event management ecosystem in Ethiopia faces several critical challenges:

1. High cost of event organization services: Most existing platforms charge high service fees, making them inaccessible for smaller organizers, individuals, or youth-led projects.
2. Lack of a client-driven event management system: There is no platform that allows users to independently plan, organize, and execute events under their own supervision. Existing systems primarily support event organizer companies rather than empowering the clients themselves.
3. Absence of coordination leading to event overlaps: Major events often occur at the same time such as concerts, exhibitions, or cultural nights, because organizers have no access to a centralized dashboard of upcoming events. This causes audience overload, venue congestion, and financial loss for both hosts and attendees.
4. Unregulated ticket sales and overbooking: Some event organizers oversell tickets for profit, leading to overcrowded venues, security risks, and poor attendee experience.

5. Lack of digital integration with government bodies: Event licensing, security approval, and other official processes remain manual and time-consuming. International and regional organizers face unnecessary delays due to the absence of a digital permit and approval workflow.

These limitations create a significant gap in Ethiopia's event ecosystem. The proposed system aims to address this gap by providing:

- An affordable, subscription-based platform for organizers
- A centralized dashboard displaying upcoming events by category
- Transparent and controlled ticketing mechanisms
- A digital channel for licensing, permissions, and government communication
- A platform where local and international organizers can plan events efficiently and securely

In summary, the project addresses the need for an accessible, coordinated, and transparent event management solution that supports both domestic and international event hosts.

1.4. Objective of the Project

1.4.1 General Objective

To design, develop, and implement an integrated digital event management platform that enables users to efficiently plan, organize, coordinate, and manage events through a unified, transparent, and user-driven system.

1.4.2. Specific Objectives

The specific objectives of the project are to:

1. Analyze and document system requirements based on the needs of event organizers, vendors, participants, and government stakeholders.
2. Design a comprehensive system architecture that supports role-based access, event planning tools, vendor management, licensing workflows, and attendee services.

3. Develop a client-driven event creation and management module that allows users to design, schedule, and supervise their events independently.
4. Implement a centralized event listing and scheduling mechanism to reduce event overlaps, promote coordination, and provide visibility of upcoming events across categories.
5. Create a vendor marketplace module where users can browse, evaluate, and book service providers based on availability, ratings, and service types.
6. Integrate a digital licensing and approval workflow that enables government bodies to review event proposals and issue permits electronically.
7. Develop a transparent and secure ticketing system that prevents overbooking and enhances the overall attendee experience.
8. Incorporate intelligent assistance or recommendation tools to support organizers in budgeting, vendor selection, and event optimization.
9. Implement analytics and reporting features to provide insights into attendance, revenue, budgeting, and overall event performance.
10. Ensure multilingual support, accessibility compliance, and usability testing to make the platform accessible to both local and international users.
11. Test, evaluate, and validate the system through functional, performance, and user acceptance testing to ensure reliability and effectiveness.

1.5 Scope and Limitation of the Project

1.5.1 Scope of the project

The scope of this project is centered on developing a web-based event management platform designed specifically for the Ethiopian event ecosystem. The system is limited to major urban areas where digital infrastructure and event-hosting activities are more developed, allowing for realistic implementation and testing. The platform supports event creation, scheduling,

vendor coordination, and regulatory communication within selected cities, ensuring that it aligns with Ethiopia's administrative processes and cultural context.

Within this scope, the project includes:

- Implementation focused on Ethiopia's major cities such as Addis Ababa, Adama, Hawassa, and other large urban centers.
- Support for event planning and scheduling relevant to Ethiopia's local practices.
- A digital environment that accommodates national regulatory workflows for event licensing and administrative communication.
- Role-based access for organizers, vendors, participants, and government representatives.
- A prototype-level system designed to function within Ethiopia's current digital infrastructure.

1.5.2 Limitation of the project

The project also operates within several boundaries imposed by time, resources, and technological feasibility. These limitations define what the system will not cover in its current development cycle. The platform does not extend to rural or semi-urban communities, nor does it attempt to integrate with international event systems, advanced real-time security tools, or large-scale commercial infrastructures.

Key limitations of the project include:

- No coverage of rural and low-infrastructure areas where digital adoption remains limited.
- No integration with foreign ticketing systems, embassy platforms, or international event-management services.
- The system remains a web-based prototype, meaning no mobile application or offline mode is included.
- High-traffic load testing, commercial scalability, and real-time crowd-management features are outside the project's scope.
- Full government system integration is simulated rather than operational.

These boundaries ensure the project remains achievable within the academic timeframe while remaining realistic for Ethiopia’s current urban digital landscape.

1.6. Deliverables

This section formally outlines the tangible outputs and products that will be delivered upon the successful completion of the Global Connect Ethiopia – International Professional Event Hub project. These deliverables are the culmination of the system development life cycle, designed to meet the specified objectives and requirements for a client-driven digital event management platform.

To clarify the value proposition, deliverables are categorized into Client-Facing Products (the system itself) and Academic & Project Artifacts (documentation required for project validation).

A. Client-Facing Product Deliverables

The core tangible output for the end-users (Organizers, Vendors, Participants, and Government).

I. Fully Functional Web Application (Global Connect Ethiopia)

- The complete, multi-user web-based platform, developed using the React, FastAPI (Python), and MongoDB stack. The application will be deployed to a public-facing hosting environment (Vercel/Render) and accessible via a registered domain.
- Client Services & Features: This platform serves as the integrated digital ecosystem for the client, delivering the following core functionalities:
 - Organizer Empowerment: A client-driven dashboard for independent event creation, design, scheduling, management, and supervision.
 - Integrated Promotional Design Utility (IPDU): A built-in tool allowing organizers to create and customize event-specific promotional materials (social media images, flyers) directly from the interface using pre-defined templates.

- Digital Marketplace: A dedicated portal for Vendors to list services, manage booking requests, and maintain profiles/ratings.
 - Government Digital Channel: A secure interface for government bodies to review event proposals and issue digital licenses/permits.
 - Centralized Coordination: A system-wide scheduling mechanism that provides visibility of upcoming events to prevent overlaps.
 - Intelligent Assistance: Automated recommendation tools to assist organizers with budgeting and vendor selection.
- Completion Criteria: Successful deployment and continuous 99% uptime during the initial testing period. All system modules must be operational, passing user acceptance testing (UAT) across the defined user roles. The IPDU must successfully generate and store customizable image files.

B. Academic & Project Artifacts

Documentation and materials required to demonstrate academic merit, technical soundness, and project feasibility.

II. Academic and Technical Reports

- Description: A comprehensive package of formal documentation:
 1. Technical System Manual: A detailed report on the system architecture, FastAPI backend structure, database schema (MongoDB), API endpoint documentation (OpenAPI), and the IPDU design logic.
 2. Final Academic Report: The complete project thesis/paper documenting the problem statement, objectives, methodology (Agile SDLC), analysis (UML), implementation, and conclusions.
- Completion Criteria: All documentation is finalized, peer-reviewed for technical accuracy, and stored in a version-controlled repository.

III. Test Data and Performance Report

- Description: A consolidated report detailing Quality Assurance (QA) activities, including unit testing, integration testing, and performance metrics (response times, load handling).
- Completion Criteria: The system successfully handles simulated traffic loads as defined in the project scope. All critical bugs identified during Sprint 5 are resolved and documented.

IV. Deployment and Maintenance Manual

- Description: A guide for system infrastructure, including instructions for re-deploying frontend/backend components, database backup/restoration (MongoDB Atlas), and API key management.
- Completion Criteria: The manual must be verified by a third party to ensure the system can be re-deployed independently.

V. Presentation and Demonstration Materials

- Description: A professional presentation package, including slides, a live demonstration script, and an executive summary video highlighting the platform's innovative features.
- Completion Criteria: Successful final project defense and demonstration to stakeholders/faculty.

1.7. Feasibility Study

1.7.1. Technical Feasibility

Technical feasibility analyzes the availability of the necessary resources, hardware, software, and technical expertise required to successfully develop and deploy the system.

Technology Stack Suitability: The project utilizes a modern, robust, and open-source technology stack consisting of React.js for the frontend, FastAPI (Python) for the backend, and MongoDB for the database. These technologies are industry standards, well-documented, and highly compatible, ensuring a stable and scalable system architecture.

Development Expertise: The development team, composed of senior Computer Science and Engineering students, possesses the requisite skills in full-stack development, database design, and system architecture. The choice of Python (FastAPI) leverages the team's strong academic foundation in algorithmic logic and data processing.

Infrastructure Availability: The system is designed to run on standard cloud infrastructure. Development requires only standard personal computers and internet access, which are readily available at Adama Science and Technology University. Deployment utilizes cost-effective platforms like Vercel and Render, which eliminate the need for expensive, proprietary on-premise servers.

Integration Capabilities: The technical design supports essential integrations, such as the Canva API for the Integrated Promotional Design Utility (IPDU) and local payment gateways (e.g., Chapa), which are technically feasible via standard RESTful API protocols.

1.7.2. Operational Feasibility

Operational feasibility examines the degree to which the system aligns with existing business processes and the likelihood of user adoption and satisfaction.

Process Improvement:

The platform addresses critical operational bottlenecks in the current manual ecosystem, such as physical ticketing, in-person licensing, and fragmented vendor searches. By digitizing these workflows, the system drastically reduces administrative overhead for both event organizers and government bodies.

User Acceptance and Usability:

To ensure high adoption rates, the system features a user-centric design with role-specific dashboards (Organizer, Vendor, Participant, Government). The interface supports both English and Amharic, lowering language barriers and ensuring accessibility for a wider demographic in Ethiopia.

Administrative Compliance:

The system respects the existing regulatory framework by digitizing the actual permit approval workflow used by local authorities, rather than attempting to bypass it. This alignment increases the likelihood of support from government stakeholders.

Maintenance:

The project uses modular code structures (Microservices/Modular Monolith via FastAPI), making long-term maintenance, debugging, and feature updates manageable for future developers or administrators.

1.7.3. Economic Feasibility

Economic feasibility assesses the cost-effectiveness of the project, weighing the development and operational costs against the projected benefits and value generated.

Cost-Benefit Analysis:

The development cost is minimal, primarily restricted to hosting fees, domain registration, and internet access (estimated at approximately 2,000 – 4,000 ETB for the initial phase). In contrast, the potential economic benefit is substantial, as the platform facilitates revenue generation for local SMEs, vendors, and event organizers who currently lack digital visibility.

Resource Utilization:

The project maximizes resource efficiency by utilizing open-source frameworks (React, FastAPI, MongoDB) that require no licensing fees. This approach significantly lowers the barrier to entry compared to purchasing proprietary event management software.

Return on Investment (ROI):

While the initial academic prototype is non-commercial, a commercial deployment would see a high ROI through a potential subscription model or commission-based ticketing fees. The platform also generates indirect economic value by boosting event tourism and professional networking within Ethiopia.

1.8. Significance of the Project

The Global Connect Ethiopia platform represents a major step toward digital transformation in Ethiopia's professional event industry. Its significance can be analyzed in four major dimensions:

A. Technological Significance

- Introduces modern web architecture integrating React (frontend), Nest.js (backend), and MongoDB (database) — providing students with hands-on experience in full-stack development.
- Demonstrates practical application of distributed computing, API integration, and cloud-based deployment.
- Promotes secure data management through encryption, role-based access, and multi-factor authentication—essential for handling VIP and corporate event data.
- Serves as a foundation for future innovations, such as AI-powered matchmaking, real-time translation, or mobile app extensions.

B. Academic and Research Significance

- Serves as a capstone application of theoretical courses (Software Engineering, Databases, Networks, and Web Security).
- Encourages problem-solving and critical thinking, as students implement real-world business logic, performance optimization, and user experience design.
- Contributes to academic literature by offering a localized case study for digital event management systems in developing countries.

C. Socio-Economic Significance

- Digitalizes Ethiopia's event sector, reducing manual coordination and increasing efficiency.
- Provides visibility for local vendors and SMEs, enabling fair competition and wider market access.

- Creates networking opportunities between local startups, investors, and global professionals.
- Supports Ethiopia’s tourism, investment, and innovation ecosystem by providing an accessible event-hosting platform.
- Encourages local tech entrepreneurship and job creation in software development, event management, and digital services.

D. Institutional and National Significance

- Aligns with the Ethiopian Digital Transformation Strategy (2025) by promoting ICT-based solutions for national industries.
- Positions Ethiopia as a regional hub for international conferences and digital collaboration.
- Contributes to the government’s goal of building a knowledge-based economy by fostering innovation and professionalism.
- Sets a model for academic–industry collaboration, showing how university projects can address real societal needs.

1.9. Beneficiaries of the Project

The proposed system provides tangible benefits to multiple groups within the Ethiopian event ecosystem. Its client-driven architecture ensures that organizers, vendors, government

officials, and attendees experience streamlined processes tailored to their needs. The platform enhances the planning experience, reduces manual overhead, and creates new opportunities for digital service delivery.

The key beneficiaries include:

- **Event Organizers:**

They gain a self-service environment to plan, schedule, and manage events without relying on external firms. The system reduces cost, improves coordination, and enhances control over the entire event lifecycle.

- **Vendors and Service Providers:**

Catering companies, photographers, designers, logistics providers, and other vendors benefit from increased visibility, digital bookings, and customer ratings that enhance trust.

- **Participants and Attendees:**

They benefit from clear schedules, secure digital ticketing, reduced overbooking, and improved access to event information.

- **Government and Regulatory Bodies:**

Authorities responsible for licensing, security clearance, and event oversight can receive digital proposals, review requests, and issue approvals through a unified interface, reducing administrative delays.

- **Corporate and International Hosts (NGOs, Embassies, Institutions):**

These stakeholders can plan and coordinate professional events in Ethiopia more efficiently, supported by standardized workflows and reliable digital communication.

- **Academic Community:**

Students, researchers, and faculty benefit from a practical, real-world system demonstrating modern software engineering practices, architecture, and user-centered design.

1.10. Methodology

The Agile Software Development Life Cycle (SDLC) was chosen for this project because it supports flexibility, iterative progress, and constant stakeholder collaboration. This approach ensures that the system can evolve through continuous user feedback and testing, producing a functional prototype at every phase.

1.10.1 Agile Development Framework

Agile development divides the project into iterative cycles (sprints), typically lasting two to four weeks. Each sprint results in a deliverable component of the system—allowing early testing, stakeholder review, and refinement.

Phase 1: Requirement Analysis (Sprint 1)

- Objective: Understand stakeholder needs and define the functional and non-functional requirements.
- Activities:
 - Conducted interviews with event organizers, vendors, and participants.
 - Identified pain points in current manual event coordination systems.
 - Drafted the Software Requirement Specification (SRS) document.
 - Created initial user personas and use cases to model system roles.
- Deliverable: SRS document and approved backlog of system features

Phase 2: System Design (Sprint 2)

- Objective: Translate requirements into a technical blueprint.
- Activities:
 - Designed system architecture (frontend, backend, and database).
 - Created UML diagrams (use case, sequence, and activity diagrams).
 - Designed the database schema in MongoDB with collections for events, users, payments, and vendors.
 - Designed wireframes and UI prototypes using Figma.
- Deliverable: System design documentation and architectural diagrams.

Phase 3: Development (Sprints 3–4)

- Objective: Build the system components based on approved designs.
- Activities:
 - Developed the frontend interface using React.js with responsive components and routing.
 - Implemented backend APIs using FastAPI, ensuring modularity and RESTful communication.
 - Integrated MongoDB Atlas for secure cloud data storage.
 - Added role-based authentication (organizer, vendor, participant, VIP).
 - Implemented payment gateway (Chapa API) and data analytics dashboard.
- Deliverable: Functional version of the Global Connect Ethiopia web platform.

Phase 4: Testing and Integration (Sprint 5)

- Objective: Ensure system functionality, performance, and security compliance.
- Activities:
 - Conducted unit and integration tests using Jest and Postman.
 - Performed User Acceptance Testing (UAT) with sample event data.
 - Evaluated system for load handling, data encryption, and VIP security features.
 - Fixed identified bugs and optimized response times.
- Deliverable: Tested, integrated, and optimized system ready for deployment.

Phase 5: Deployment and Documentation (Sprint 6)

- Objective: Deploy the system for public access and prepare documentation.
- Activities:
 - Hosted frontend on Vercel and backend on Render, linked with MongoDB Atlas.
 - Registered the custom domain for public access.
 - Prepared user manuals, technical documentation, and final presentation slides.
- Deliverable: Fully functional deployed system, documentation, and project presentation.

Phase 6: Maintenance and Feedback

- Objective: Continuously improve the system based on user feedback.

- Activities:
 - Monitor user behavior and system analytics.
 - Collect feedback from early adopters (organizers, vendors).
 - Implement updates and performance improvements through periodic sprints.
- Deliverable: Updated and scalable version of the platform ensuring long-term usability.

1.11 Development tools

Table 1 : Development Tools

Category	Tool / Technology	Purpose / Usage
Frontend Technologies	React.js	Builds dynamic, modular, and responsive user interfaces.
	Tailwind CSS / Material UI	Provides fast styling, layout structure, and reusable UI components.
Backend Technologies	FastAPI (Python)	Develops high-performance backend APIs with built-in validation and documentation.
	Uvicorn	ASGI server used to run the FastAPI backend locally.
Database Tools	MongoDB Community Edition	Local NoSQL database for storing events, users, vendors, tickets, and system records.
	MongoDB Compass	GUI for managing collections, running queries, and monitoring local database activity.
Development & Design Tools	Figma / Adobe XD	Used for UI/UX wireframing, prototyping, and interface design.

	VS Code	Primary integrated development environment for coding and debugging.
	IPDU (Integrated Promotional Design Utility)	Custom tool for generating event promotional materials within the system.
Testing Tools	Postman	Tests backend API endpoints through localhost.
	PyTest	Performs unit and integration tests for backend components.
	Jest + React Testing Library	Tests frontend components, UI behavior, and interactions.
	Browser Developer Tools	Debugs client-side scripts, layout, and performance issues.
Project Management Tools	Git & GitHub	Manages version control, pull requests, team collaboration, and repo hosting.
	Trello / Jira	Supports sprint planning, task tracking, and Agile workflow management.
Local Deployment Tools	Node.js + NPM	Runs the React frontend and manages dependencies.
	Python Virtual Environment (venv)	Manages isolated backend dependency environments.
	Docker & Docker Compose (optional)	Containerizes the full stack for consistent local deployment.

1.12. Required Resources with Cost

Table 2 : Required Resources with Cost

Resource	Description	Cost (ETB)
Laptops	Developer machines (owned)	—
Internet	Campus lab	-
Domain & Hosting	Web domain and deployment	1 000
Documentation	For printing	1000
Total	-	2000

1.13. Task and Schedule.

Table 3 : Task and Schedule

Project Phase	Tasks / Activities	Timeline
Phase 1: Documentation & Planning	Problem analysis, requirement gathering, drafting Chapters 1–3, literature review, methodology, UML draft, advisor revisions, final documentation submission.	Until December 29
Phase 2: System Design	System architecture, database schema, UI/UX wireframes, technical specification preparation.	Early Jan – Mid Jan
Phase 3: Implementation	Front-end (React), backend (FastAPI), MongoDB setup, authentication, event workflows, vendor modules, dashboards, initial IPDU development.	Jan – Apr (4 months)
Phase 4: Testing & Integration	Unit testing, integration testing, bug fixes, performance improvement, user acceptance testing.	Late April
Phase 5: Final Presentation & Submission	Demo preparation (localhost), submission of final documentation & code, presentation, project defense.	Early May

1.14. Team Composition

Table 4 :Team Composition

Team Member	Assigned Role	Primary Responsibilities
Biniyam Getachew	Database Developer	● Design MongoDB schema for users, events, vendors, tickets. ● Configure local MongoDB and manage collections. ● Create indexes and optimize queries. ● Prepare seed data for testing and demo.
Samrawit Alemu	Backend Developer	● Develop RESTful APIs using FastAPI. ● Implement authentication, authorization, and business logic. ● Integrate backend with MongoDB. ● Manage local backend server setup and deployment environment.
Meron Gedamu	UI/UX Designer	● Create wireframes, mockups, and prototypes using Figma. ● Design user flows and interface layouts. ● Prepare visual guidelines for frontend implementation. ● Support usability testing and design revisions.
Mikiyas Assefa	Frontend Developer	● Build user interfaces using React.js. ● Apply responsive styling using Tailwind CSS. ● Integrate frontend with backend APIs.

		● Handle client-side state management and UI logic.
Miraf Tsegaye	QA Engineer	● Prepare test cases for UI, API, and database operations. ● Conduct unit tests (PyTest, Jest) and integration testing. ● Identify, document, and track bugs. ● Lead user acceptance testing and final system validation.

CHAPTER 2 : DESCRIPTION OF EXISTING SYSTEM AND LITERATURE REVIEW

2.1. Major Function of Existing System

Existing event management systems, both within Ethiopia and across global platforms, provide essential digital services that support the planning, coordination, and execution of events. These systems have evolved to address fundamental challenges such as ticket sales, event promotion, attendee registration, vendor management, and analytical reporting. Although the level of sophistication varies between local and international tools, the core objective remains consistent: enabling organizers to streamline event operations efficiently.

In the Ethiopian context, current systems include local platforms such as Tikva Events, Shengo Events, and several homegrown event-organizing companies that maintain partial digital infrastructures. Local ticketing and event-aggregation platforms, including Ahun App and Chapa Tickets, are also notable contributors to the ecosystem. These systems typically provide basic functions such as event listing, ticket purchase, promotional support, and attendee registration. Their strength lies in contextual relevance—integrating local payment methods, addressing cultural expectations, and supporting promotional strategies suited to the Ethiopian market. However, they often rely on manual or semi-digital workflows, lacking

fully automated vendor-booking modules, advanced analytics, or large-scale event scheduling capabilities.

In contrast, global reference systems such as Eventbrite, Ticketmaster, Cvent, Meetup, and Eventzilla offer highly developed event-management environments. These platforms support end-to-end workflows, beginning with event creation and extending to ticket sales, attendee management, session scheduling, on-site check-in, and post-event data analysis. They incorporate marketing tools such as automated email campaigns, discount code management, and integrated promotional dashboards. Advanced systems further provide vendor-booking capabilities, virtual event support, exhibitor management, and communication modules that streamline collaboration between organizers, participants, and stakeholders.

The major functions of these systems can be understood through the essential tasks they perform. Ticket sales remain a primary feature, enabling organizers to define ticket types, set prices, control availability, and process secure payments. Event promotion functions include publishing event pages, generating promotional materials, and enabling built-in social sharing. Vendor-booking features, available in more sophisticated platforms, allow organizers to contract service providers such as caterers, photographers, decorators, or technical teams directly through the system. Event scheduling functions assist in defining timelines, session structures, locations, and resource allocation. Basic registration processes are facilitated through attendee sign-ups, digital confirmation, and automated reminders. Marketing tools improve audience reach, while analytics modules offer insights into attendance patterns, ticket sales performance, and user engagement.

The operational workflow of these systems typically begins with user input. Organizers submit essential event information, including event details, ticket configurations, payment preferences, promotional materials, and additional registration requirements. The system processes these inputs by publishing event pages, managing ticket inventory, facilitating financial transactions, and storing registration data securely. During this process, the platform automates communication through confirmation messages, event reminders, and updates. On event day, many systems support attendee check-in using QR codes or digital tickets. Finally, organizers receive comprehensive outputs such as revenue reports, attendance records, and performance analytics that inform future event planning.

Overall, existing event management systems play a crucial role in modernizing the event industry by integrating administrative, promotional, and analytical functions into unified digital platforms. Their capabilities range from basic listing and ticketing services found in local Ethiopian systems to comprehensive, data-driven, and fully automated solutions offered by global platforms. Understanding these existing systems provides a foundation for identifying gaps and designing improved, localized solutions tailored to the Ethiopian context.

2.2 Users of the Current System

Existing event-management systems are used by multiple stakeholder groups whose needs, expectations, and level of satisfaction vary depending on the sophistication of the platform and the local context in which it operates. In Ethiopia, where the digital event industry is still developing, these users interact with systems in ways that reflect both emerging digital habits and persistent limitations in infrastructure. Understanding the characteristics, motivations, and experiences of each user group is essential for evaluating the shortcomings of current systems and guiding the design of improved solutions.

Event Organizers

Event organizers are the primary users of event-management systems. This group includes professional planners, entertainment promoters, cultural-event coordinators, youth-led initiatives, private companies, and independent organizers. Their central need is to efficiently plan, promote, and manage events through a centralized digital environment. They typically expect tools for ticket sales, event promotion, audience management, and real-time data on registrations. Organizers in Ethiopia, for instance, often rely on partially digital systems or a combination of social media, mobile-money services, and third-party ticketing platforms. While these tools help increase visibility and streamline payment collection, organizers frequently encounter gaps such as the absence of built-in scheduling tools, limited vendor-management capabilities, and inadequate analytics. As a result, many operational tasks continue to be handled manually, reducing scalability and increasing the risk of miscommunication.

Vendors and Service Providers

Vendors including:

- Caterers
- Decorators
- Audiovisual Technicians
- Photographers
- Security providers
- Venue operators

form another important user group. Their interest in event-management systems lies primarily in access to clients. Vendors expect exposure to event organizers, the ability to showcase services, and clear communication about event requirements. In global platforms, vendors can participate in integrated marketplaces that allow them to receive bookings, manage availability, and build credibility through ratings. In Ethiopia, however, a formalized digital vendor ecosystem is largely absent. Most vendor interactions take place through personal networks or direct messaging. This lack of structured support reduces market transparency, limits opportunities for small vendors, and creates inefficiencies in event planning. Vendors often express the need for a digital system that reliably connects them with clients, standardizes communication, and prevents last-minute logistical conflicts.

Participants and Attendees

Participants who attend events, purchase tickets, or register for sessions form the largest numerical user base. Their primary needs include easy registration, secure payment, reliable event information, and a smooth check-in process. Attendees expect seamless digital experiences that provide clarity regarding schedules, pricing, venue directions, and updates. In Ethiopia, platforms that support digital ticketing have greatly improved access, but users still report challenges such as limited payment options, inconsistent event information, and unreliable confirmation systems. These issues can lead to mistrust, especially for high-demand events where overselling or misinformation has occurred in the past. Attendees increasingly expect digital confirmations, QR-code tickets, and timely notifications all of which are inconsistently available in current systems.

Corporate Hosts and Institutions

Corporate hosts including NGOs, embassies, educational institutions, private companies, and government-linked organizations use event systems for conferences, seminars, workshops, and professional gatherings. Their needs extend beyond basic ticketing and include advanced registration systems, participant categorization (e.g., speakers, VIPs), official communications, and post-event reporting. Globally, systems such as Cvent or Eventzilla provide such functionality through detailed analytics, customizable registration forms, and automated communication workflows. In Ethiopia, corporate hosts often resort to manual tools like spreadsheets and email lists, supplemented by basic registration modules from local platforms. They frequently find that existing systems lack the level of professionalism, data management, and security required for formal or diplomatic events. This gap reduces operational efficiency and complicates stakeholder coordination.

Government and Licensing Bodies

Although government offices play a central role in regulating events through licensing and compliance checks, their direct use of event-management systems remains minimal. In many cases, application processes for permits, security approvals, and venue clearance remain largely manual. While government entities stand to benefit from digital integration—such as automated verification of event details or centralized event scheduling—the absence of such systems creates delays for organizers. Licensing bodies expect accuracy, transparency, and reliable documentation, but current systems rarely provide structured interfaces for official review. This disconnect between event platforms and regulatory processes represents a major limitation in Ethiopia’s event ecosystem.

Summary of User Expectations and Limitations

Across all user categories, common expectations include efficiency, transparency, and reliability. Event organizers seek comprehensive tools that reduce manual workload; vendors want structured access to clients; participants expect convenience and trustworthy information; institutions need professional features and analytics; and government bodies require verifiable data and streamlined workflows. However, existing systems—particularly in Ethiopia—do not fully meet these expectations due to fragmented services, limited automation, and lack of integrated vendor and regulatory functions.

The gaps experienced by these users highlight the need for an improved event-management platform that centralizes activities, enhances communication, supports decision-making through analytics, and aligns digital functions with real-world processes. Understanding the users of current systems forms a crucial foundation for designing solutions that address the unique challenges and opportunities of Ethiopia's event ecosystem.

2.3 Drawback of the Current System

Event management in Ethiopia is still limited by fragmented workflows, outdated tools, and the absence of integrated digital platforms. Existing local and international systems fail to support the full event lifecycle or meet Ethiopia's regulatory and infrastructural needs. The following major drawbacks justify the need for a unified event hub like our project.

1. Limited to Ticketing and Registration Only

Most event systems in Ethiopia only provide basic ticketing and attendee registration services. They do not include critical tools such as event planning, budgeting, vendor coordination, resource allocation, or on-site management. Because these features are missing, clients cannot design or control their events independently. Instead, they must depend heavily on event organizing companies that handle everything manually, which reduces efficiency, increases mistakes, and makes the whole process slow and expensive.

2. No Client-Led Event Planning Tools

Current platforms do not allow clients whether individuals, businesses, or international participants to plan events by themselves. Features like drag-and-drop layout design, automated budgeting, venue suggestions, scheduling tools, and resource management are completely missing. This lack of self-service tools forces clients to rely on organizers even for simple tasks, reduces transparency, and increases the overall cost and time needed to prepare an event.

3. No Integration With Government Licensing and Approvals

There is no digital connection between event systems and government offices responsible for permissions, security approval, or licensing. Organizers still have to visit offices physically, submit paper documents, make phone calls, and follow long manual workflows. This causes

delays, miscommunication, and an increased risk of events happening without proper authorization. The absence of a unified system also makes it difficult for authorities to monitor upcoming events and ensure compliance with safety regulations.

4. Event Scheduling Conflicts

Since Ethiopia lacks a centralized event coordination dashboard, it is common for major events, concerts, conferences, exhibitions, and festivals to overlap on the same day. This leads to audience splitting, venue competition, parking issues, and reduced ticket sales for organizers. Without a national event calendar or conflict-detection feature, organizers have no visibility into what other events are happening, resulting in poor planning and unnecessary financial losses.

5. Overbooking and Weak Control Systems

Many events experience overbooking, fake ticket duplication, and poor crowd control because the current systems do not provide real-time validation or attendee tracking. Tickets can be easily copied or oversold beyond venue limits, creating safety concerns and damaging the reputation of event organizers. Without digital gate systems, QR validation, or live attendance dashboards, organizers have very little control over entrance management.

6. No Vendor Marketplace

There is no system that connects event organizers or clients to reliable vendors such as caterers, decorators, sound engineers, DJs, photographers, security services, or transportation companies. People must depend on personal contacts or random social media posts, which often leads to inconsistent quality and delays. A missing vendor marketplace means clients cannot compare prices, check ratings, or communicate with multiple providers in a structured way.

7. International Systems Are Not Localized

International platforms like Eventbrite, Hopin, and Cvent are not suitable for Ethiopia because they are not designed for local payment methods, do not support government licensing processes, and lack Amharic or other local language support. These systems assume fast internet speeds and stable infrastructure, which many Ethiopian regions do not have.

Additionally, they are expensive and primarily built for Western markets, making them inaccessible for most local event organizers.

8. No Multi-Role Dashboards

Event organizers, vendors, government authorities, and attendees all require different dashboards to manage their responsibilities. However, current systems treat all users the same and offer no specialized interfaces. Without role-based dashboards, communication becomes chaotic, approvals are delayed, and event workflow management becomes complex. This lack of separation also reduces accountability and transparency among stakeholders.

9. High Costs and Limited Automation

Because most event processes are manual such as budgeting, vendor coordination, pricing, and reporting, event management becomes expensive and time-consuming. There is no automation for generating quotations, scheduling tasks, monitoring finances, or analyzing attendee behavior. This results in inefficiency, inconsistent pricing, and higher service costs for clients.

2.4 Literature Review

This section reviews scholarly work related to digital event management, digital transformation, e-government automation, marketplace platforms, ticketing and crowd systems, smart-city technologies, ICT adoption in Ethiopia, and user-centered design. The goal is to provide academic grounding for the Global Connect Ethiopia project and identify existing gaps that the system aims to address.

1. Digital Event Systems

Research shows that event management has increasingly shifted from manual processes to digital platforms that integrate registration, ticketing, scheduling, and communication. Ryan (2020) studied the “digital maturity” of events and found that organizations progress through stages—from basic online registrations to fully integrated platforms that support analytics, audience engagement, and real-time updates. Higher digital maturity correlates with improved attendee experiences and operational efficiency.

Prototype studies such as those published in *Procedia Computer Science* (2020) demonstrate that most web-based event management tools focus on simple functions like event creation, attendee lists, and basic notifications. However, many lack advanced modules such as marketplace features, role-based access, vendor onboarding, or analytics dashboards.

These studies indicate that while event-management systems exist, many are feature-limited and not designed for multi-stakeholder ecosystems like vendors, organizers, VIPs, and participants—an area directly addressed by Global Connect Ethiopia.

2. Digital Transformation in Event Planning

Digital transformation research consistently highlights the shift from traditional operations to technology-enabled, data-driven processes. Verhoef et al. (2021) argue that transformation requires integration across systems, customer-centric design, and continuous improvement rather than simply digitizing old workflows.

Specifically within the event sector, Ragab and Salama (2021) explain that digital solutions improve planning efficiency, sustainability, and coordination among event stakeholders. Their work shows that digital tools—such as automated registration, centralized vendor information and integrated communication—are critical for modern event management.

This literature reinforces the need for a unified digital platform that supports multiple event processes rather than separate tools, which aligns with the proposed Global Connect Ethiopia system architecture.

3. E-Government Integration and Service Automation

E-government studies, particularly within Ethiopia, provide insights into user adoption, institutional readiness, and challenges in digital service delivery. Studies show that user trust, perceived usefulness, and system quality are major determinants of e-government adoption. Poor digital literacy, low internet quality, and lack of awareness remain obstacles. AAU studies (2021) further report that citizens' readiness to adopt digital services improves with training, clarity of information, and well-designed user interfaces.

These findings indicate that any digital platform designed for the Ethiopian public must prioritize trust, accessibility, local language support, and a simple user experience. Global

Connect Ethiopia incorporates these considerations by offering Amharic/English interfaces, transparent vendor information, and secure workflows for VIP/government events.

4. Vendor–Client Digital Marketplaces

Marketplace literature focuses on how platforms connect service providers and customers. Hagiu and Wright (2020) describe multi-sided platforms as systems where value increases with active participation from both vendors and users. Their findings stress that platform policies, incentives, and trust mechanisms shape vendor behavior and user satisfaction. Shi et al. (2021) showed that promotion strategies, such as featured listings and recommendation algorithms, increase marketplace engagement.

Research on e-commerce trust models (2022) further reveals that clear vendor profiles, transparent communication, verified reviews, and secure payment options increase adoption.

These findings directly inform the vendor directory and marketplace components of Global Connect Ethiopia, which include verified vendor profiles, rating systems, and secure digital payments (e.g., Chapa integration).

5. Ticketing and Crowd Management Systems

Crowd management is a critical component of large event platforms. Still (2014) provides foundational principles for crowd behavior, emphasizing the need for controlled entry, pre-event planning, and monitoring of attendee flow. Safety Science studies (2019) show that congestion risks can be reduced through time-slot ticketing, dynamic admission, and real-time monitoring. IEEE research on QR and RFID-based ticketing (2020) demonstrates the effectiveness of digital ticketing for secure access control.

These findings justify the integration of QR-generated tickets, slot-based scheduling, and live capacity dashboards in Global Connect Ethiopia.

6. Smart-City Digital Platforms

Smart-city research discusses the integration of digital tools to support urban services. IEEE Smart Cities Initiative (2019) outlines architectures that emphasize modularity, interoperability, and real-time data sharing. Talari et al. (2017) identify common challenges such as system fragmentation and lack of centralized platforms. Studies on digital urban

services (2020–2023) highlight the role of integrated platforms in enhancing city-level coordination, especially for events, transportation, and public communication.

These insights support the platform’s long-term goal of integrating with city services, mapping, government systems, and event-infrastructure data.

7. ICT Adoption in Ethiopia and Developing Countries

Studies on ICT adoption in Ethiopia (Misganaw, 2020; Belachew, 2021) reveal recurring barriers: inconsistent connectivity, limited digital skills, trust issues, and the high cost of technology. Public-sector ICT diffusion research (AAU, 2020–2022) also notes that successful adoption depends on awareness, usability, affordability, and training initiatives.

This emphasizes the importance of designing systems with low-bandwidth support, intuitive workflows, and strong trust-building mechanisms. Global Connect Ethiopia addresses these issues through optimized UI, Amharic/English options, secure registration, and simplified workflows for both vendors and organizers.

8. User-Centered Design in Digital Event Platforms

User-centered design (UCD) is essential for ensuring accessibility and adoption. Norman (2013) argues that systems must reduce cognitive load, minimize steps, and offer intuitive interactions. Abras et al. (2004) describe UCD as an iterative process that relies on user feedback, prototyping, and usability testing.

These principles guide the design of Global Connect Ethiopia, ensuring that interfaces are simple, visually clear, multilingual, and role-specific.

Gap Analysis

The literature demonstrates strong global progress in event technology, digital transformation, e-government integration, and marketplace platforms. However, several gaps remain—especially in developing-country contexts such as Ethiopia:

1. Lack of integrated systems:

Existing event platforms focus on single functions like ticketing or registration.

2. **Limited local adaptation:**

Most systems reviewed are designed for high-income countries. They do not reflect Ethiopia's ICT challenges, multilingual needs, or local payment systems.

3. **Missing vendor–client ecosystems:**

Research on marketplaces highlights the importance of trust and transparency, yet most event prototypes lack built-in vendor directories, ratings, or marketplace modules.

4. **Incomplete crowd management features:**

While studies recommend time-slot ticketing and real-time capacity monitoring, these features are rarely found in systems used in the Ethiopian context.

5. **Weak focus on user experience in developing countries:**

Studies indicate that digital literacy and trust affect adoption, but local event platforms do not consistently prioritize UX, onboarding, or accessibility.

Global Connect Ethiopia fills these gaps by providing a comprehensive, multilingual, secure, and user-centered digital platform designed specifically for Ethiopia's event ecosystem. It unifies event scheduling, ticketing, vendor marketplace services, VIP modules, and analytics into a single scalable system that aligns with both global research and local needs.

Chapter 3: Proposed System

3.1. Overview

This chapter focuses on defining and analyzing the system's requirements, beginning with an outline of both functional and non-functional specifications. To demonstrate how the proposed solution addresses current issues, it presents various operational scenarios. The chapter concludes by providing a comprehensive set of UML-compatible models, including use-case, activity, sequence, and class diagrams, along with data persistence details.

3.2. Functional requirement

Table 5: Functional Requirements

ID	Functionality
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FR-1	The system shall allow organizers to submit event proposals with required documents for government approval.
FR-2	The system shall allow government and municipal officers to review, approve, or reject proposals and issue digital event permits.
FR-3	The system shall allow organizers to create events, add event details, and publish them to a public calendar.
FR-4	The system shall allow vendors to register, upload legal documents, and be approved by administrators.
FR-5	The system shall allow organizers to request quotes, negotiate, and sign digital contracts with vendors.
FR-6	The system shall allow organizers to set ticket types, and attendees shall register or purchase tickets online.
FR-7	The system shall process online ticket payments through integrated payment gateways and generate digital tickets.
FR-8	The system shall allow hotels to list available accommodations or resources for events, and allow users to book them.
FR-9	The system shall generate QR badges and provide checklists and incident reporting tools for on-site event management.

FR-10	The system shall collect attendee feedback and generate summary reports for organizers.
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This section specifies the functional requirements of the proposed system or solution, such as the actions, behaviors, or operations that it must perform or support, or the services or outcomes that it must provide or produce.

3.3. Non-functional requirement

Table 6: Non-Functional Requirements

ID	Functionalities
NFR1 — Performance	Core pages shall load within 2 seconds under normal network conditions.
NFR2 — Availability	The system shall maintain at least 99.5% uptime.
NFR3 — Security	All sensitive data shall be encrypted in transit and at rest using industry-standard protocols.
NFR4 — Access Control	Role-based access control (RBAC) shall be enforced for organizers, vendors, government offices, and administrators.

NFR5 — Scalability	The system shall scale to support large events with high concurrent users during ticketing peaks.
NFR6 — Usability	The interface shall be simple and allow new users to create an event within 10 minutes.
NFR7 — Multi-Language Support	The platform shall support English and Amharic throughout major user flows.
NFR8 — Compatibility	The system shall function properly on major browsers and mobile devices.
NFR9 — Reliability & Backup	System data shall be backed up daily and have a recovery point objective (RPO) of 24 hours.

3.4. System Model

3.4.1 Scenario

Scenario 1: User Registration and Account Creation

1. The user accesses the Global Connect Ethiopia platform through a web browser
2. The user selects the “Register” option.
3. The system displays the registration form.

4. The user enters personal details such as name, email, role selection (organizer, vendor, attendee), and password.
5. The system validates the input data.
6. The system creates a new user account and stores it in the database.
7. The user receives a confirmation message and is redirected to the login page or dashboard.

Scenario 2: User Login and Role-Based Access

1. The user opens the Global Connect Ethiopia platform.
2. The user enters valid login credentials.
3. The system verifies the credentials.
4. The system identifies the user's role.
5. The system redirects the user to the appropriate dashboard based on their role.
6. The user gains access to authorized system features.

Scenario 3: Organizer Requests Vendor Services

1. The organizer logs into the system.
2. The organizer navigates to the vendor marketplace section.
3. The system displays a list of approved vendors.
4. The organizer views vendor profiles and service details.
5. The organizer sends a service request or quotation request to selected vendors.
6. The system notifies the vendors of the request.
7. Vendors respond with quotes or availability information.

Scenario 4: Vendor Responds to Organizer Request

1. The vendor logs into the system.

2. The vendor views incoming service requests.
3. The vendor reviews event details and service requirements.
4. The vendor submits a quotation or proposal.
5. The system delivers the response to the organizer.
6. The organizer reviews and proceeds with negotiation or agreement.

Scenario 5: Attendee Registers for an Event

1. The attendee accesses the public event listing.
2. The attendee selects an event of interest.
3. The system displays event details and ticket options.
4. The attendee selects a ticket type.
5. The attendee completes registration or payment.
6. The system confirms the registration and issues a digital ticket.

Scenario 6: Event Check-in and Attendance Verification

1. The attendee arrives at the event venue.
2. The attendee presents a QR code ticket.
3. The organizer or event staff scans the QR code using the system.
4. The system validates the ticket.
5. The system marks the attendee as checked-in.
6. The organizer views real-time attendance updates.

Scenario 7: Feedback Collection After Event

1. The event concludes.
2. The system sends feedback requests to attendees.

3. Attendees submit ratings and comments.
4. The system stores feedback data.
5. The system generates summary reports for the organizer.

3.4.2. Use Case Description

Table 7: Submit Event Proposal

UC-01	Submit Event Proposal
Actor	Organizer
Precondition	The organizer is registered, logged into the system, and has the required event and legal documents.
Postcondition	The event proposal is successfully submitted and stored with a “Pending Review” status.
Normal Flow	1. Organizer logs into Global Connect Hub. 2. Organizer navigates to “Submit Proposal” and fills required fields (event title, type, dates, venue request, expected attendees, draft program, budget summary, security plan). 3. Organizer uploads required documents (PDF): organizer registration, insurance, draft contract. 4. Organizer clicks Submit; system generates a Proposal ID and shows confirmation. 5. System routes proposal to Ministry queue for review.

Alternative Flow	Organizer saves as Draft and returns later to submit.
Exception Flow	• File upload fails or required field missing → system blocks submit and shows validation errors. • if repeated upload failure, system logs error and sends alert to admin.

Table 8:Ministry Review & Verification

UC-02	Ministry Review & Issue Verification Letter
Actor	Government Reviewer (Ministry)
Precondition	A submitted event proposal exists and the ministry officer is authenticated.
Postcondition	The proposal review decision is recorded and the organizer is notified.
Normal Flow	1. Reviewer opens assigned proposal from ministry dashboard. 2. The reviewer inspects documents and metadata. 3. The reviewer selects the Approve → system, generates a digitally signed Verification Letter and forwards to the City Municipal queue. 4. The system notifies the Organizer of verification and provides a letter download link.

Alternative Flow	Reviewer requests changes → system sends a “Revision Request” to Organizer with a checklist; proposal status becomes <i>Requires Revision</i> .
Exception Flow	Reviewer rejects due to non-compliance → system marks proposal <i>Rejected</i> and emails Organizer with rejection reason.

Table 9: *Municipal Location Allowance & Police Notification*

UC-03	Municipal Location Allowance & Police Notification
Actor	Municipal Officer
Precondition	The proposal has passed the initial ministry review.
Postcondition	Municipal verification status is updated and forwarded to the next approval stage.
Normal Flow	1. Municipal Officer receives ministry-verified proposal. 2. The officer checks the local calendar, venue suitability, and safety considerations. 3. If acceptable, the officer issues a Location Allowance and submits a security request to the nearest police station through the platform. 4. System updates permit status to <i>Municipal Approved</i>.

Alternative Flow	Municipal Officer proposes alternate dates/locations → sends proposal back to Organizer via system.
Exception Flow	Police station denies feasibility due to conflict or capacity → municipal sets status <i>Denied</i> and notifies Organizer.

Table 10: create & publish event

UC-04	Create & Publish Event (Organizer)
Actor	Organizer
Precondition	The event proposal is approved by all required authorities.
Postcondition	The event is created and published on the public event listing.
Normal Flow	1. Organizer selects Create Event. 2. Completes event metadata (title, description, capacity, location [linked to municipal approval], program schedule, VIP list) and selects ticketing mode. 3. Organizer presses Publish; system validates required approvals (if government event requires permit the system checks permit status).

	4. The event becomes publicly listed on the Centralized Calendar.
Alternative Flow	The organizer publishes as Private (invite-only) and shares invite links.
Exception Flow	Required permit is missing or rejected → publishing is blocked and Organizer is notified.

Table 11: Reserve Venue

UC-05	Reserve Venue
Actor	Organizer
Precondition	An approved event exists and venue data is available.
Postcondition	The selected venue is reserved and linked to the event
Normal Flow	1. Organizer queries venue availability for desired dates. 2. The system shows availability and cost estimates. 3. Organizer sends reservation request; venue provider confirms via platform. 4. Reservation status = <i>Confirmed</i> and linked to the event.

Alternative Flow:	Venue offers alternate dates or partial availability; Organizer accepts alternate.
Exception Flow	Venue declines or double-booked → reservation fails; system suggests alternate venues.

Table 12: Vendor Registration

UC-06	Vendor Registration
Actor	Vendor Owner
Precondition	The vendor has access to the platform and valid registration documents.
Postcondition	The vendor application is submitted for administrative approval.
Normal Flow	1. Vendor selects Register as Vendor and completes business profile: business name, TIN, trade license, service categories, sample portfolio, contact details. 2. Vendor uploads TIN copy and business registration scan. 3. Vendor submits; system marks status <i>Pending Approval</i>.

Alternative Flow	Vendor saves partial profile as draft and finishes later.
Exception Flow	Missing mandatory legal document → system rejects registration and provides checklist.

Table 13: Approve vendors

UC-07	Approve Vendor
Actor	Admin
Precondition	Vendor registration requests exist and the administrator is authenticated.
Postcondition	The vendor is approved or rejected and notified accordingly.
Normal Flow	1. Admin reviews pending vendor registration documents. 2. Admin verifies TIN and business registration fields. 3. Admin marks Approve; system activates vendor accounts and publishes their service listing.

Alternative Flow	Admin requests additional documents or clarifications → vendor notified to upload.
Exception Flow	Documents fraudulent or invalid → Admin rejects vendor and records the decision (audit log).

Table 14: Vendor List Service & Quote Request

UC-08	Vendor List Service & Quote Request
Actor	Organizer , Vendor
Precondition	Approved vendors exist and the organizer is logged in
Postcondition	Quote requests are sent to selected vendors.
Normal Flow	1. Organizer browses vendor marketplace and selects service (e.g., catering). 2. The organizer requests a quote by submitting event details and guest count. 3. Vendor receives request and submits a quote through platform. 4. Organizer reviews and accepts quotes; booking status = <i>Booked</i>.

Alternative Flow	Vendor replies with counter-offer → Organizer can accept, negotiate, or reject.
Exception Flow	Vendor fails to respond within SLA → system escalates to alternate vendors and notifies Organizer.

Table 15: Create & Sign Contract

UC-09	Create & Sign Contract (Record Payment Terms)
Actor	Organizer & Vendor
Precondition	Vendor quotes have been accepted by both parties.
Postcondition	A digital contract is created, signed, and stored.
Normal Flow	1. After quote acceptance, the platform generates a draft contract with milestones. 2. Vendor and Organizer review and e-sign (electronic signature). 3. Signed contract stored and payment milestones scheduled.
Alternative Flow	Parties renegotiate terms before signing → contract updated.

Exception Flow	One party refuses to sign → booking is canceled and status updated.
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Table 16: Publish Ticketed Event & Configure Inventory

UC-10	Publish Ticketed Event & Configure Inventory
Actor	Organizer
Precondition	The event is published and ticketing is enabled.
Postcondition	Ticket inventory is configured and ticket sales are activated.
Normal Flow	1. The organizer configures ticket types (General, VIP), quantities, pricing, and seat allocation. 2. The organizer links payment options and enables ticket sales. 3. The system enforces inventory and displays remaining tickets on the event page.
Alternative Flow	Organizer sets hold-back inventory for VIP/press.
Exception Flow	Invalid price/capacity inputs → validation error; publishing blocked.

Table 17: Purchase Paid Ticket

UC-11	Purchase Paid Ticket
Actor	Attendee
Precondition	Tickets are available and the attendee has access to payment services.
Postcondition	Payment is completed and a digital ticket is issued.
Normal Flow	1. Attendee selects ticket type and proceeds to checkout. 2. Attendee completes payment via integrated gateway (local supported gateway). 3. Payment gateway returns success; system issues QR ticket and confirmation email.
Alternative Flow	Attendee selects offline payment method (bank transfer) and receives provisional booking pending confirmation.
Exception Flow	• Payment fails (card declined or timeout) → transaction aborted, seat released • user shown failure message and retry option.

Table 18: Prevent Overbooking

UC-12	Prevent Overbooking
Actor	System
Precondition	Ticket inventory limits are defined.
Postcondition	Ticket sales stop automatically when capacity is reached.
Normal Flow	1. The system uses transactional locking or optimistic concurrency when finalizing purchases. 2. The first successful transaction decrements inventory and issues tickets; simultaneous attempts handled in order.
Alternative Flow	If queueing is enabled, users are placed in a short reservation hold.
Exception Flow	Race condition or DB error → system rolls back and displays sold-out; incident logged for admin review.

Table 19: Publish Accommodation Vacancy

UC-13	Publish Accommodation Vacancy
Actor	Hotel Manager
Precondition	Accommodation provider is registered and approved.
Postcondition	Accommodation listings become visible for booking.
Normal Flow	1. Host creates vacancy listing (dates, price, capacity, location) tied to specific event. 2. The platform verifies host identity and publishes listings. 3. Vendors search and book listed vacancies.
Alternative Flow:	Host marks listing Blocked for VIP allocations.
Exception Flow	Host fails identity verification → listing not published and host notified.

Table 20: Reserve Accommodation

UC-14	Reserve Accommodation
Actor	Vendors

Precondition	Accommodation listings are available and the user is authenticated.
Postcondition	The accommodation booking is confirmed and recorded.
Normal Flow	1. Vendors select vacancy and request booking. 2. Host confirms booking and optionally requires deposit. 3. Booking confirmed 4. Calendar updated.
Alternative Flow	The host offers alternate dates or an upgraded room.
Exception Flow	Double-booking or payment failure → booking canceled and attendee notified.

Table 21: Generate & Print Badge

UC-15	Generate QR Check-in and badge
Actor	Organizer
Precondition	The attendee has a valid ticket and completed registration.
Postcondition	A badge with QR code is generated and ready for printing.
Normal Flow	1. Organizer exports attendee list for a session or all attendees.

	2. The system generates printable badges with QR codes and roles (Attendee/Vendor/VIP). 3. On-site staff print and distribute badges during check-in.
Alternative Flow	Use on-site badge printing service via vendor booking.
Exception Flow	Badge generation failure (font/layout error) → system offers CSV export fallback.

Table 22: On-site Incident Report

UC-16	On-site Incident Report
Actor	On-site Staff
Precondition	An active event exists and the user has reporting permission.
Postcondition	The incident report is logged and accessible to authorities.
Normal Flow	1. Staff logs incidents (type, time, description, photos) into the incident module. 2. System notifies Organizer and, if severe, municipal/police contacts per permit settings. 3. Incident status tracked until resolved.

Alternative Flow	Minor incidents are logged and closed by staff without escalation.
Exception Flow	Notification delivery fails → system retries and logs failure for admin follow-up.

Table 23: Send Post-Event Survey & Collect Feedback

UC-17	Send Post-Event Survey & Collect Feedback
Actor	System
Precondition	The event has concluded and attendees exist
Postcondition	Feedback responses are collected and stored.
Normal Flow	1. After Event Completion, the system schedules and sends a feedback survey link to attendees. 2. Attendees submit responses; system aggregates and scores feedback. 3. The organizer receives summarized reports including NPS, satisfaction, vendor ratings.

Alternative Flow	Send segmented surveys (VIPs / vendors / general attendees) with custom questions.
Exception Flow	Low response rate → system triggers reminder emails; if sending fails, logs error for admin.

Table 24: Publish Final Event Roadmap & Lessons Learned

UC-18	Publish Final Event Roadmap & Lessons Learned
Actor	Organizer
Precondition	Event feedback and reports are available.
Postcondition	Final documentation is published for internal reference.
Normal Flow	1. The organizer composes the final roadmap (timeline, costs, vendors used, lessons learned) in the dashboard. 2. The organizer sets visibility (private to sponsors/government or public). 3. System archives and notifies selected stakeholders; analytics stored for benchmarking.

Alternative Flow	Organizer requests platform anonymized benchmarking report vs similar past events.
Exception Flow	Document upload or export fails → system provides raw JSON export as fallback.

Table 25: Generate Event Schedule via AI

UC-19	Generate Event Schedule via AI
Actor	Organizer , AI Engine
Precondition	Event details and session data exist.
Postcondition	An AI-generated draft schedule is produced.
Normal Flow	1. Organizer selects "AI Schedule Assistant" within the event dashboard. 2. Organizer inputs constraints: Event Type (e.g., Conference), Duration (2 days), and Start Time. 3. The system sends prompts to LLM with context. 4. The system returns a suggested hour-by-hour itinerary (Keynotes, Breaks, Networking). 5. Organizer reviews and clicks "Apply to Calendar."

Alternative Flow	The organizer manually edits specific time slots before applying.
Exception Flow	AI service timeout → System displays manual schedule entry form.

Table 26: Generate Event Schedule via AI

UC-20	Chatbot Inquiry for Licensing Rules
Actor	Organizer (International/Local)
Precondition	Registered user asking a question
Postcondition	Licensing guidance is displayed to the user.
Normal Flow	1. The organizer opens the "Help & Regulations" Chatbot. 2. Organizer types query (e.g., "Do I need a police permit for 500 people?"). 3. System queries the regulatory database/LLM. 4. Chatbot responds with specific Ethiopian licensing rules and links to the correct Proposal Form.
Alternative Flow	Chatbot cannot answer → offers a button to "Contact Ministry Official."

Exception Flow	Service unavailable → displays static FAQ page.

Table 29: Invite Committee Member

UC-21	Invite Team Member
Actor	Organizer
Precondition	The event exists and the organizer is authenticated.
Postcondition	Committee invitations are sent to selected users
Normal Flow	1. Lead Organizer navigates to "Event Team" settings. 2. Enters email address of the new staff member. 3. Assigns specific roles (e.g., "Finance Manager" or "Check-in Staff"). 4. The system sends invitation links via email.

	5. Invitee accepts; system links them to the Event Dashboard.
Alternative Flow	The organizer revokes an invitation before it is accepted.
Exception Flow	User already exists in team → System shows "User already assigned" error.

Table 30: Assign Tasks to Committee

UC-22	Assign Tasks to Team
Actor	Organizer
Precondition	Committee members have accepted invitations.
Postcondition	Tasks are assigned and tracked in the system.
Normal Flow	1. The organizer creates a task (e.g., "Finalize Catering Menu"). 2. The organizer selects a team member from the dropdown list. 3. Sets a due date and priority. 4. The system notifies the assigned member.
Alternative Flow	The organizer assigns tasks to "Self."

Exception Flow	Assignee account inactive → System blocks assignment.

Table 31: Broadcast Event Announcement

UC-23	Broadcast Event Announcement
Actor	Organizer
Precondition	The event is published and recipients exist.
Postcondition	Announcements are delivered via selected channels.
Normal Flow	1. The organizer selects "Send Announcement." 2. Selects audience segment (e.g., "All Ticket Holders" or "VIP Only"). 3. Drafts message (Subject, Body) or selects a template. 4. Clicks "Send"; system dispatches emails/SMS/In-App notifications.
Alternative Flow	Organizer schedules announcement for a future date/time.

Exception Flow	Zero recipients found (no tickets sold) → System warns user before sending.
----------------	---

Table 32: Create/Edit Event Budget

UC-24	Create/Edit Event Budget
Actor	Organizer
Precondition	The organizer has an active event.
Postcondition	Budget details are saved and updated.
Normal Flow	1. Organizer accesses "Budgeting Tool." 2. Adds a line item (e.g., "Venue Rental") and estimated cost. 3. System updates the "Total Estimated Cost" visualization. 4. The organizer saves the budget version.
Alternative Flow	The organizer imports the budget from a CSV template.

Exception Flow	Negative values entered → System validation error.
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Table 33: Track Revenue Analytics

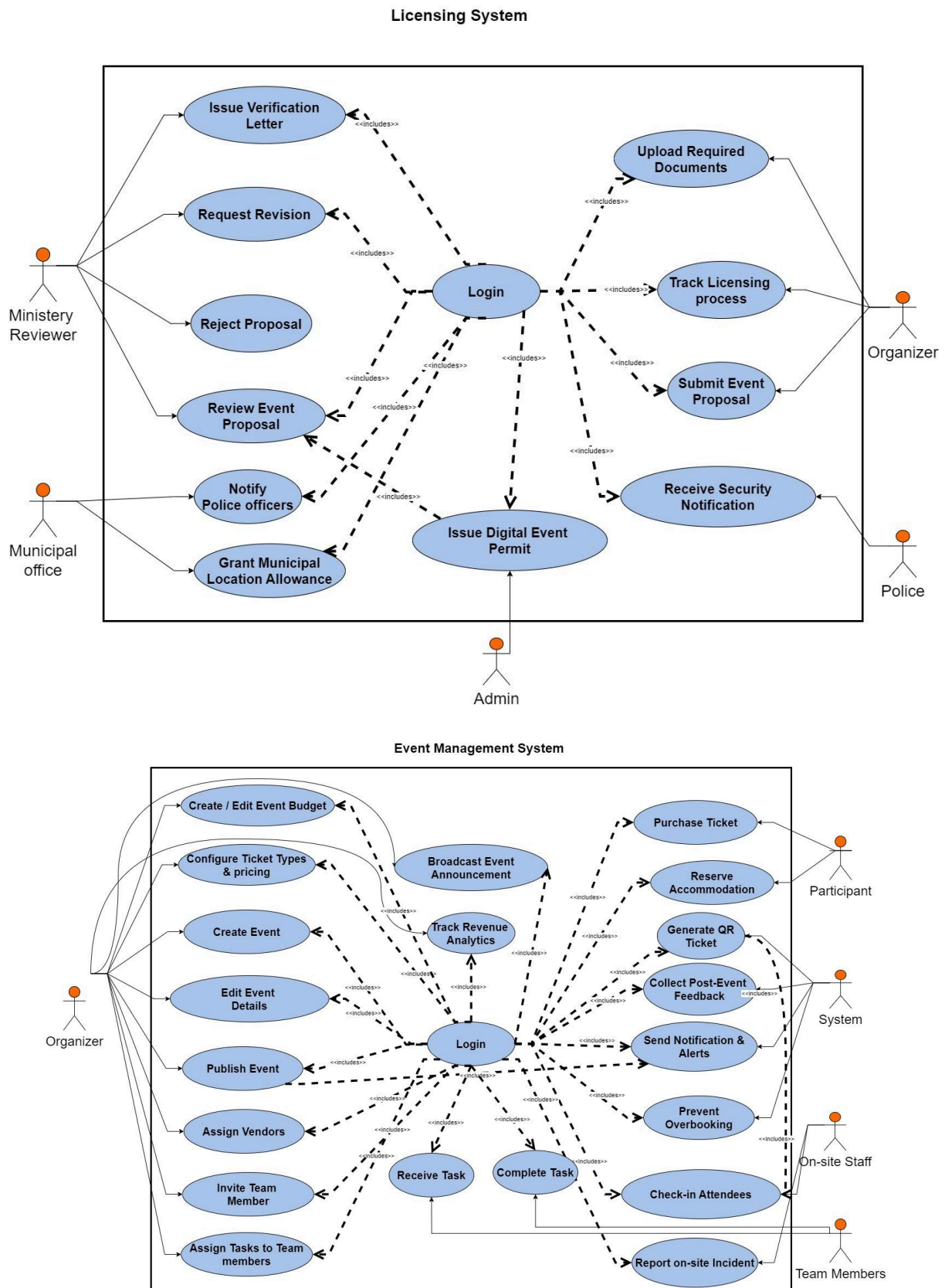
UC-25	Track Revenue Analytics
Actor	Organizer
Precondition	Financial transactions exist for the event.
Postcondition	Revenue analytics are displayed to the organizer.
Normal Flow	1. Organizer opens "Analytics Dashboard." 2. Selects "Revenue View." 3. The system aggregates ticket sales, vendor fees, and sponsorship funds. 4. The system displays a real-time graph of Revenue vs. Budget.
Alternative Flow	Organizer filters view by "Payment Method" (e.g., Telebirr vs. Bank Transfer).
Exception Flow	Data synchronization delay → System shows "Last updated: [Timestamp]" warning.

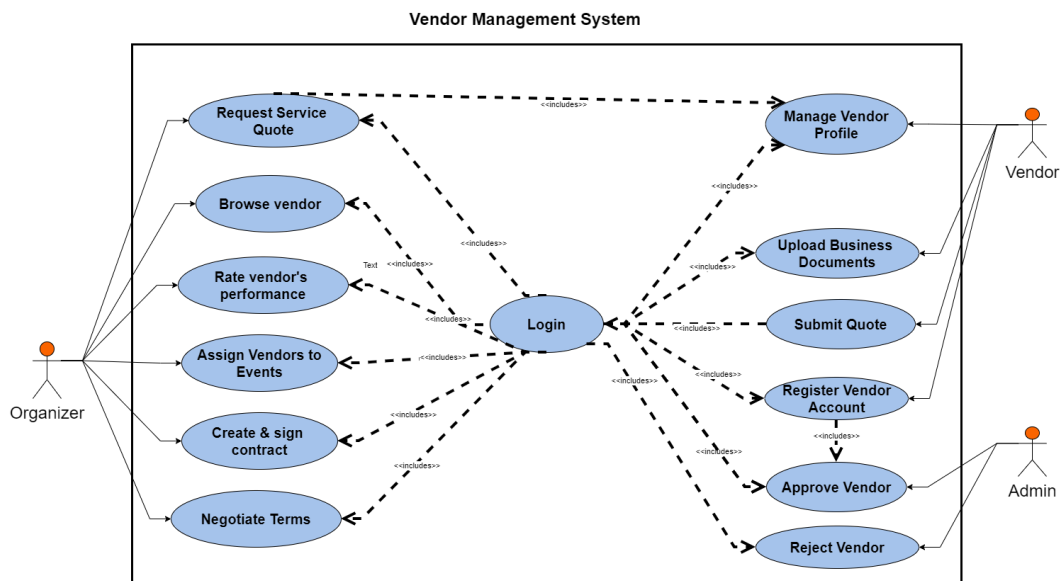
Table 34: Configure Notification Preferences

UC-26	Configure Notification Preferences
Actor	User (Organizer/Attendee)
Precondition	The user is authenticated.
Postcondition	Notification settings are saved and applied.
Normal Flow	1. User navigates to Account Settings > Notifications. 2. User toggles preferences for specific triggers (e.g., "New Ticket Sale," "Event Reminder," "Permit Status Change"). 3. The user selects channels (Email, SMS, In-App). 4. The system saves preferences.
Alternative Flow	The user selects "Unsubscribe All."
Exception Flow	Database update fails → User notified "Settings not saved."

3.4.3 Use case model

figure 1: Use-case diagram





3.5. Object Model

3.5.1. Data Dictionary

Table 9 : Data dictionary

Class	Attribute	Data Type	Constraints	Description
User	user_id	UUID	PK	Unique identifier for every user
	password_hash	String		Securely hashed password
	role	Enum	Organizer, Vendor, Attendee, Admin	User role in the platform
	status	Enum	Active, Inactive, Banned	Account status

	created_at	DateTime		Registration timestamp
	full_name	String		Public display name
	email	String	Unique	Verified email address
	phone	String		Contact phone number
	isVerified	Boolean		Email/phone verification status
UserSession	session_id	UUID	PK	Active login session ID
	user_id	UUID	FK → User	References User
	expires_at	DateTime		Session expiry timestamp
UserProfile	user_id	UUID	PK, FK → User	One-to-one profile
	bio	Text		Public biography
	organization	String		Company or affiliation
EventProposal	proposal_id	UUID	PK	Unique proposal ID
	organizer_id	UUID	FK → User	Who submitted the proposal
	status	Enum	Draft, Submitted, UnderReview...	Current review status

	budget_estimate	Decimal		Estimated total budget
	title	String		Public proposal title
	event_type	String		Conference, Concert, etc.
	start_date	Date		Proposed start date
	end_date	Date		Proposed end date
	expected_attendees	Integer		Expected number of attendees
ProposalReview	review_id	UUID	PK	Review record ID
	proposal_id	UUID	FK → EventProposal	Which proposal is reviewed
	reviewer_id	UUID	FK → User	Government/admin reviewer
	decision	Enum	Approved, Rejected, Changes	Final decision
	comments	Text		Reviewer comments
Permit	permit_id	UUID	PK	Official permit ID
	proposal_id	UUID	FK → EventProposal	Linked proposal

	city_status	Enum		City/municipal approval status
	police_status	Enum		Police/security clearance status
	issued_at	DateTime		Issuance timestamp
Document	document_id	UUID	PK	Stored file ID
	owner_id	UUID	FK → User	Uploader
	file_url	String		Cloud storage URL
	type	Enum	ProposalDoc, Permit, Contract...	Document category
Event	event_id	UUID	PK	Live event ID
	proposal_id	UUID	FK → EventProposal	Origin proposal
	organizer_id	UUID	FK → User	Event owner
	venue_id	UUID	FK → Venue	Booked venue
	status	Enum	Planned, Published...	Event lifecycle status
	title	String		Public event name
	description	Text		Full event description
	capacity	Integer		Maximum attendees
	schedule_json	JSON		Detailed timetable

Venue	venue_id	UUID	PK	Venue identifier
	provider_id	UUID	FK → User/Vendor	Owner or manager
	name	String		Venue name
	capacity	Integer		Max persons
	availability	JSON		Calendar of free slots
	location	String		Address
Reservation	reservation_id	UUID	PK	Venue booking record
	event_id	UUID	FK → Event	Booked event
	venue_id	UUID	FK → Venue	Reserved venue
	status	Enum		Confirmed, Cancelled, etc.
Vendor	vendor_id	UUID	PK	Vendor company ID
	user_id	UUID	FK → User	Linked user account
	tin	String		Tax Identification Number
	status	Enum	Pending, Approved, Suspended	Approval status
	business_name	String		Public company name
	rating	Decimal(2, 1)		Average rating (0.0–5.0)

VendorService	service_id	UUID	PK	Service offering ID
	vendor_id	UUID	FK → Vendor	Provider
	category	String		Catering, Sound, Decoration...
	price_range	String		e.g., “5000–15000 ETB”
	description	Text		Service details
Quote	quote_id	UUID	PK	Quote identifier
	vendor_id	UUID	FK → Vendor	Quoting vendor
	event_id	UUID	FK → Event	Target event
	amount	Decimal		Total quoted amount
	status	Enum	Sent, Accepted, Expired	Quote lifecycle
Contract	contract_id	UUID	PK	Signed contract ID
	quote_id	UUID	FK → Quote	Origin quote
	signed_url	String		e-Signed PDF link
	status	Enum	Draft, Signed, Terminated	Contract state
TicketType	ticket_type_id	UUID	PK	Ticket tier ID
	event_id	UUID	FK → Event	Belongs to event
	price	Decimal		Price per ticket

	quantity	Integer		Total available tickets
	name	String		VIP, General, Early Bird...
	description	Text		Tier benefits
	isActive	Boolean		Sales enabled/disabled
TicketInventory	inventory_id	UUID	PK	Inventory tracking record
	ticket_type_id	UUID	FK → TicketType	Linked ticket type
	reserved	Integer		Temporarily held tickets
	sold	Integer		Confirmed sold tickets
TicketOrder	order_id	UUID	PK	Purchase order ID
	user_id	UUID	FK → User	Buyer
	ticket_type_id	UUID	FK → TicketType	Purchased tier
	quantity	Integer		Number of tickets
	status	Enum	Pending, Paid, Refunded	Order state
	total_amount	Decimal		Final amount paid
Payment	payment_id	UUID	PK	Transaction record

	order_id	UUID	FK → TicketOrder	Linked order
	method	Enum	Card, MobileMoney, Bank	Payment method
	amount	Decimal		Paid amount
	status	Enum	Success, Failed, Pending	Payment outcome
AccommodationListing	listing_id	UUID	PK	Accommodation offer ID
	host_id	UUID	FK → User	Host
	event_id	UUID	FK → Event	Linked event
	price	Decimal		Price per night
	capacity	Integer		Beds/rooms available
	status	Enum		Available, Booked, Cancelled
	title	String		Listing title
	amenities	String		Wi-Fi, Breakfast, etc.
AccommodationBooking	booking_id	UUID	PK	Booking record
	listing_id	UUID	FK → AccommodationListing	Booked listing
	user_id	UUID	FK → User	Guest
	start_date	Date		Check-in

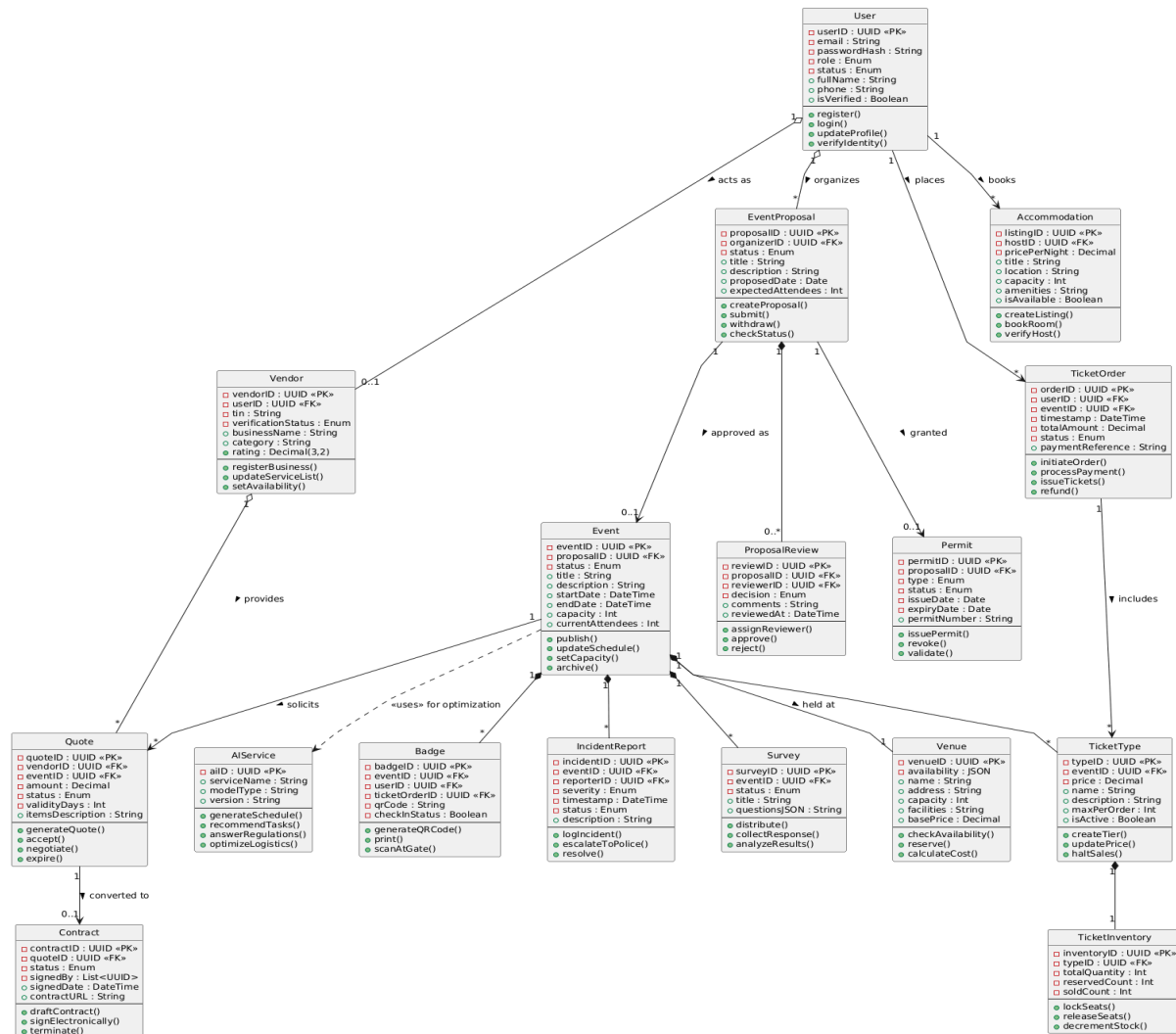
	end_date	Date		Check-out
	status	Enum		Confirmed, Cancelled, Completed
Badge	badge_id	UUID	PK	Digital/physical badge ID
	event_id	UUID	FK → Event	Event access
	user_id	UUID	FK → User	Owner
	role	Enum	Attendee, Staff, VIP, Speaker	Access level
	qr_code	String		Scannable QR code
	checked_in	Boolean		Check-in status
IncidentReport	incident_id	UUID	PK	Incident record
	event_id	UUID	FK → Event	Affected event
	reporter_id	UUID	FK → User	Who reported
	severity	Enum	Low, Medium, High, Critical	Severity level
	description	Text		Incident details
	timestamp	DateTime		When reported
Survey	survey_id	UUID	PK	Post-event survey
	event_id	UUID	FK → Event	Linked event
	title	String		Survey name
	questions	JSON		Questions structure

SurveyResponse	response_id	UUID	PK	Individual response
	survey_id	UUID	FK → Survey	Which survey
	user_id	UUID	FK → User	Respondent
	answers	JSON		Answers in JSON format
	submitted_at	DateTime		Submission time
TeamTask	task_id	UUID	PK	Organizer task
	event_id	UUID	FK → Event	Event context
	assigned_to	UUID	FK → User	Responsible person
	due_date	Date		Deadline
	status	Enum		ToDo, InProgress, Done
	title	String		Task name
EventAnalytics	analytics_id	UUID	PK	Analytics snapshot
	event_id	UUID	FK → Event	Target event
	metrics	JSON		Attendance, revenue, feedback...
	generated_at	DateTime		Report timestamp
AIService	ai_id	UUID	PK	AI assistant interaction
	event_id	UUID	FK → Event	Context event

	mode	Enum	Schedule, Budget, Risk, Summary	AI mode used
	input	Text		User/AI prompt
	output	Text		AI-generated response
	created_at	DateTime		Interaction timestamp

3.5.2. Class diagram

figure 2 : Class Diagram



3.5.3. Dynamic Model

The system manages the complete lifecycle of international and local event organization by coordinating licensing, vendor engagement, ticketing, accommodation, and on-site operations.

Organizers dynamically interact with government offices to submit proposals, receive verification letters, and obtain municipal approvals, with the system updating proposal states based on reviewer actions and regulatory responses.

Vendors and service providers respond to quote requests, sign digital contracts, and manage reservations through real-time workflow transitions. Ticketing processes include inventory locking, payment validation, and automated QR code generation. Attendees interact with event listings, accommodation options, and announcements while the system handles state-based transitions such as booking confirmation, cancellation, or expiry.

AI components assist organizers by generating automated event schedules, answering regulatory questions, and optimizing resource allocation. Collaboration features allow event teams to coordinate through task assignments, status updates, and invitation workflows. On-site operations such as check-ins, badge generation, and incident reporting follow event-driven transitions to ensure safety and efficient crowd management.

Notifications keep all stakeholders informed about approvals, contract updates, ticket sales, schedule changes, and post-event survey requests. The system continuously adapts through these dynamic interactions, ensuring a streamlined and compliant event management experience.

3.5.4. Sequence diagram

figure 3: User-login

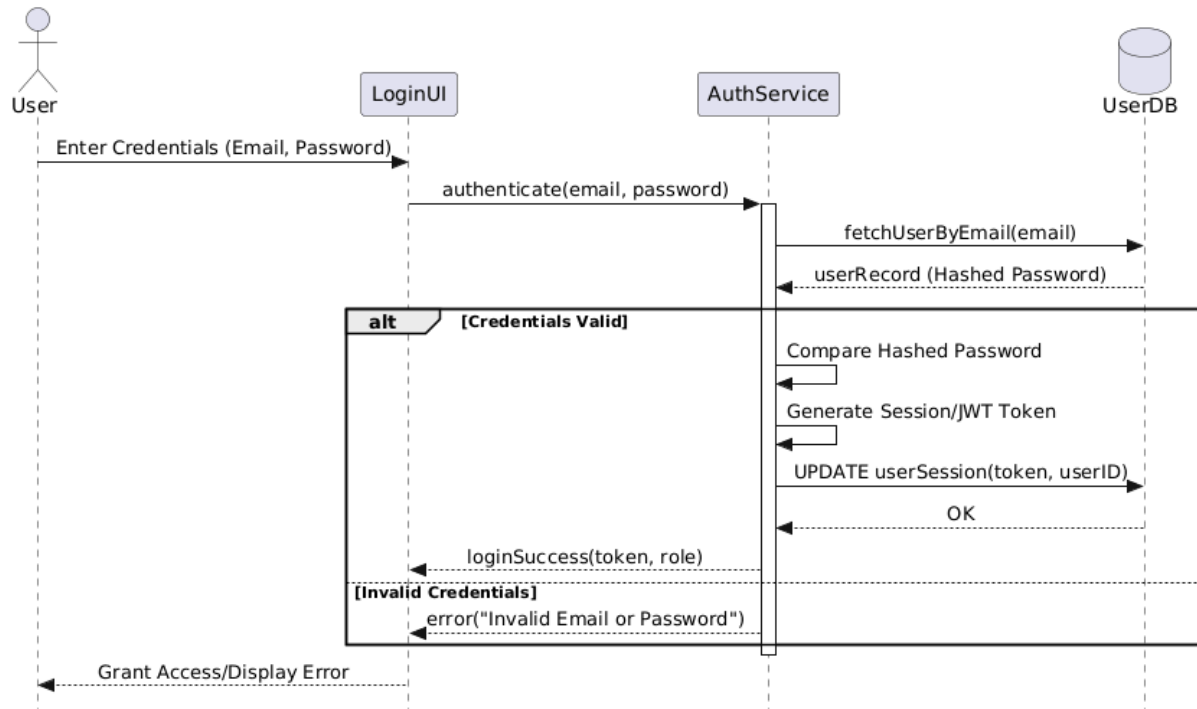


figure 4: User-registration

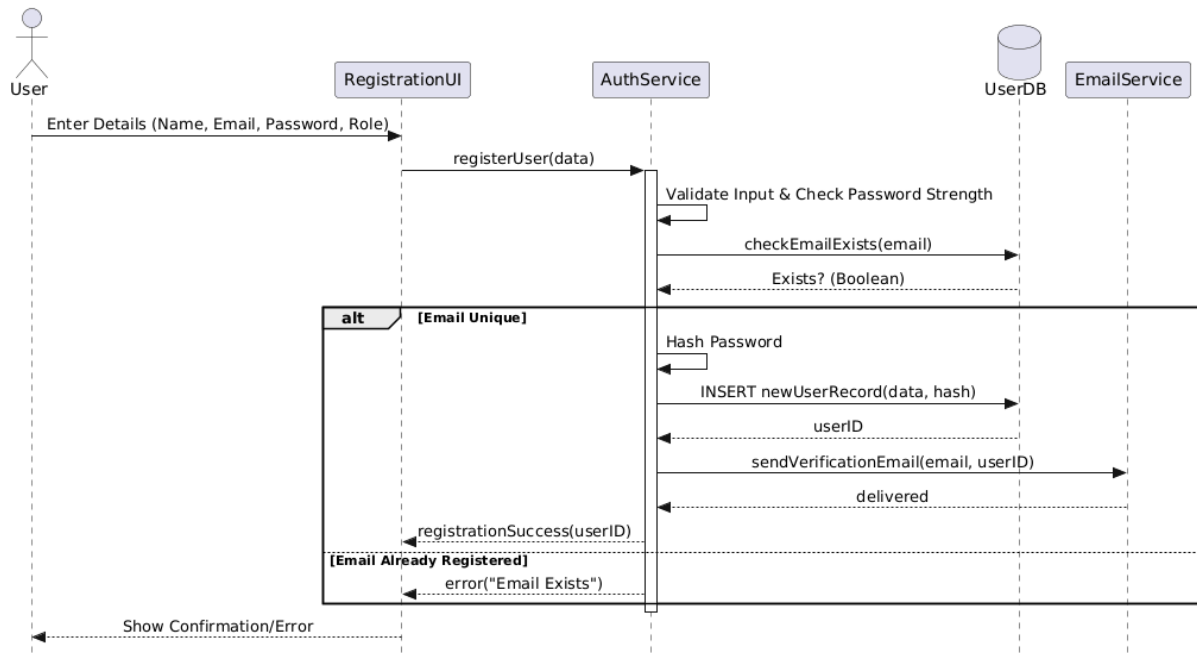


figure 5: Submit event proposal

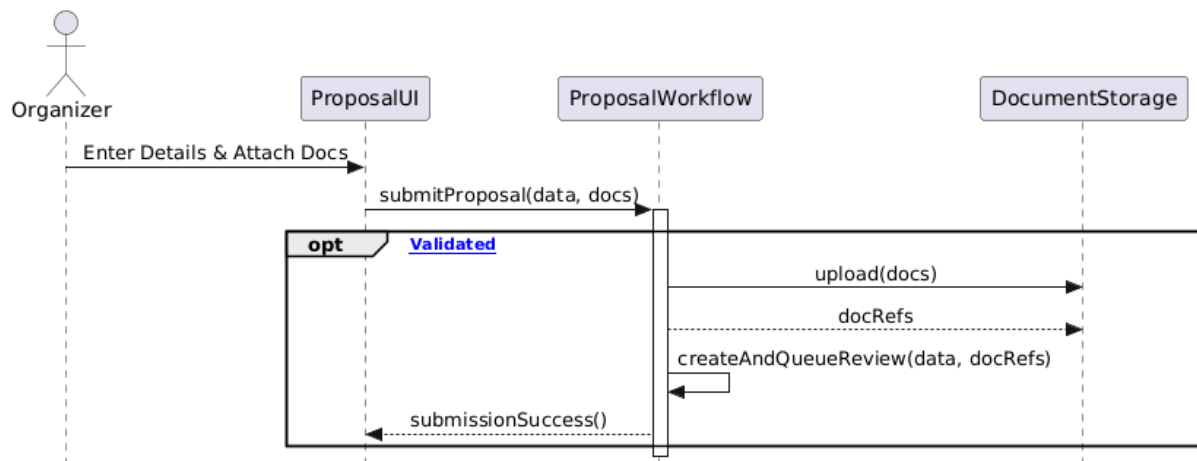


figure 6 : Ministry review & Verification

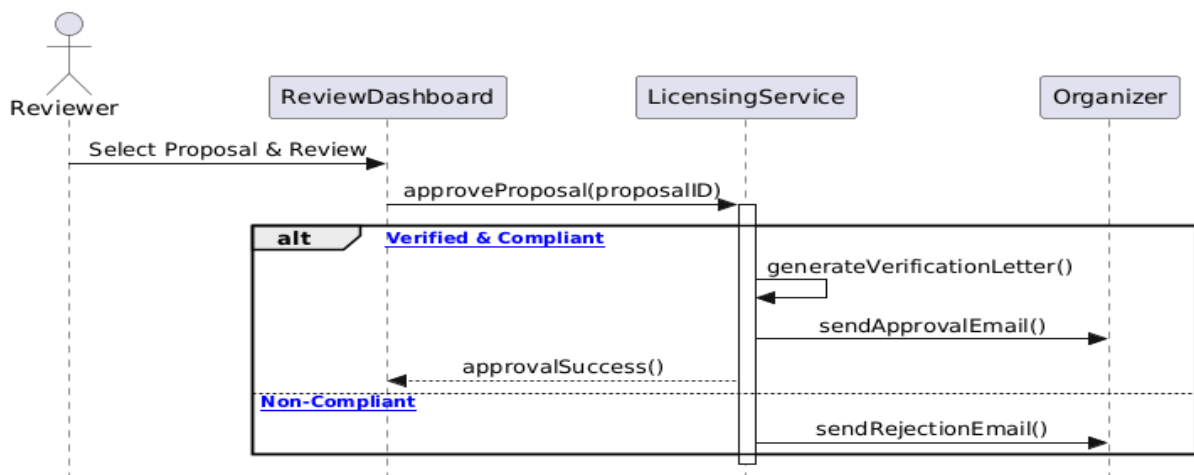


figure 7: Create and Publish event

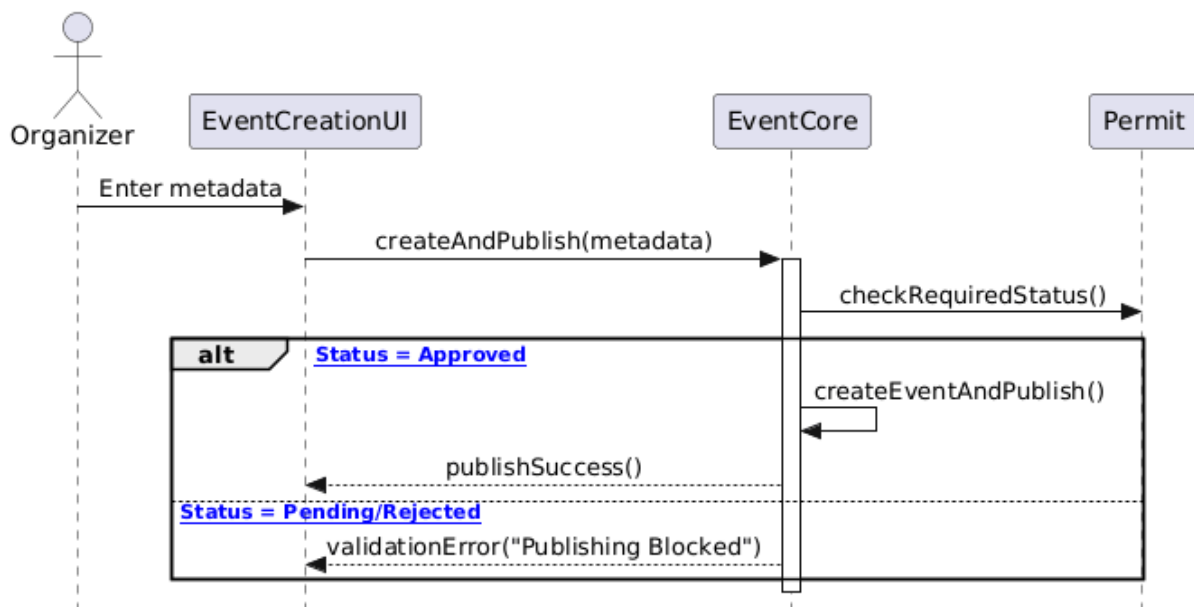


figure 8: Generate event schedule via AI

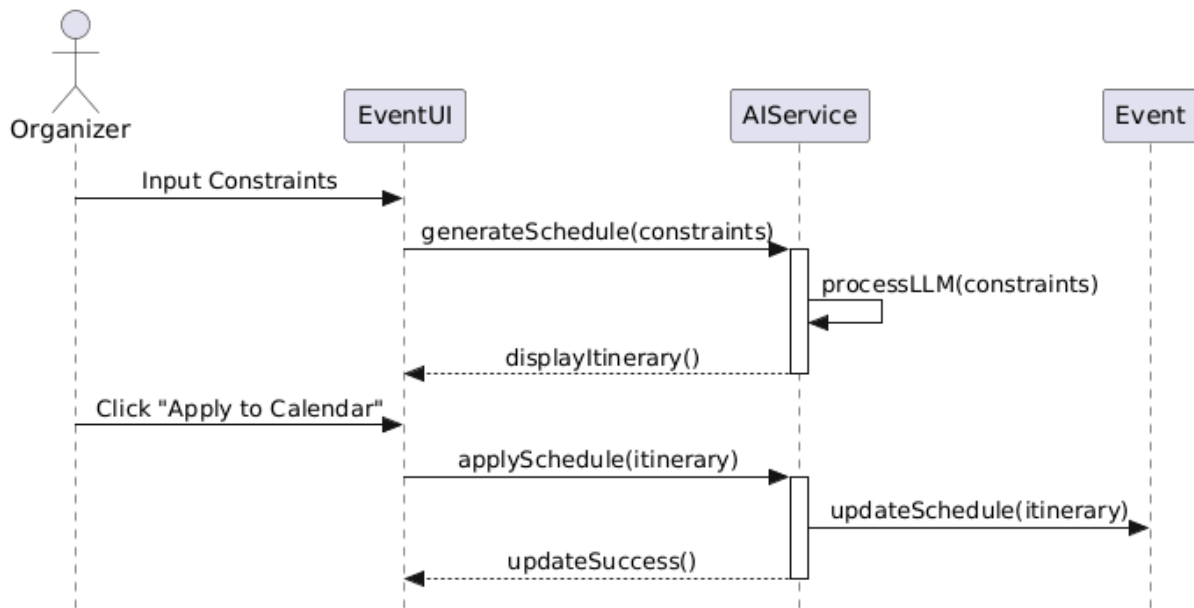


figure 9: Vendor Registration

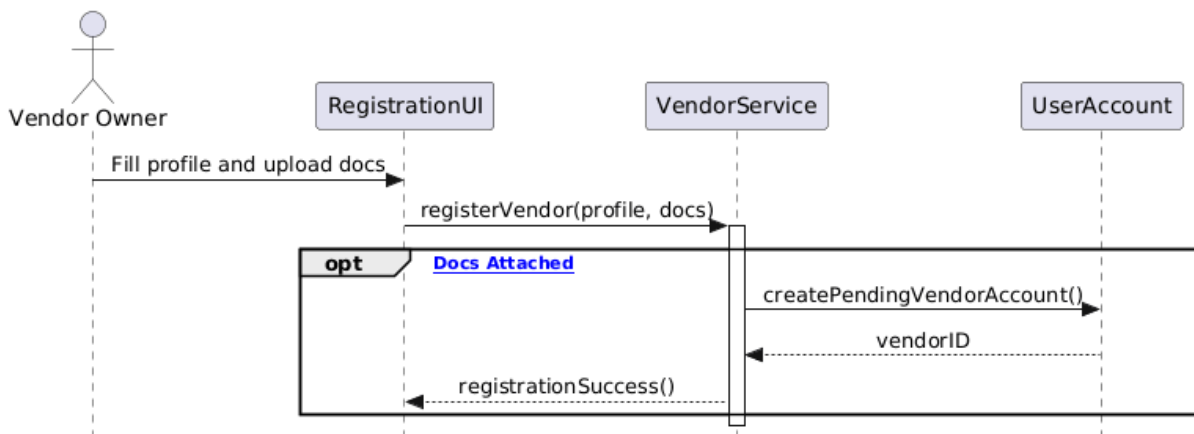


figure 10: Approve Vendor

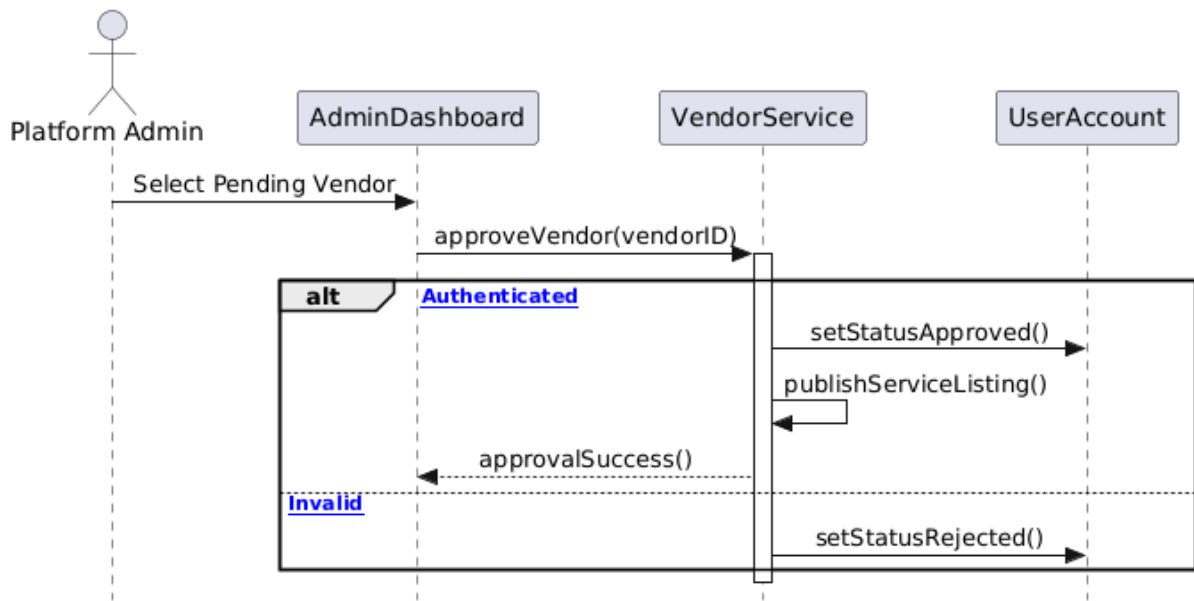


figure 11: Create ad Sign contract

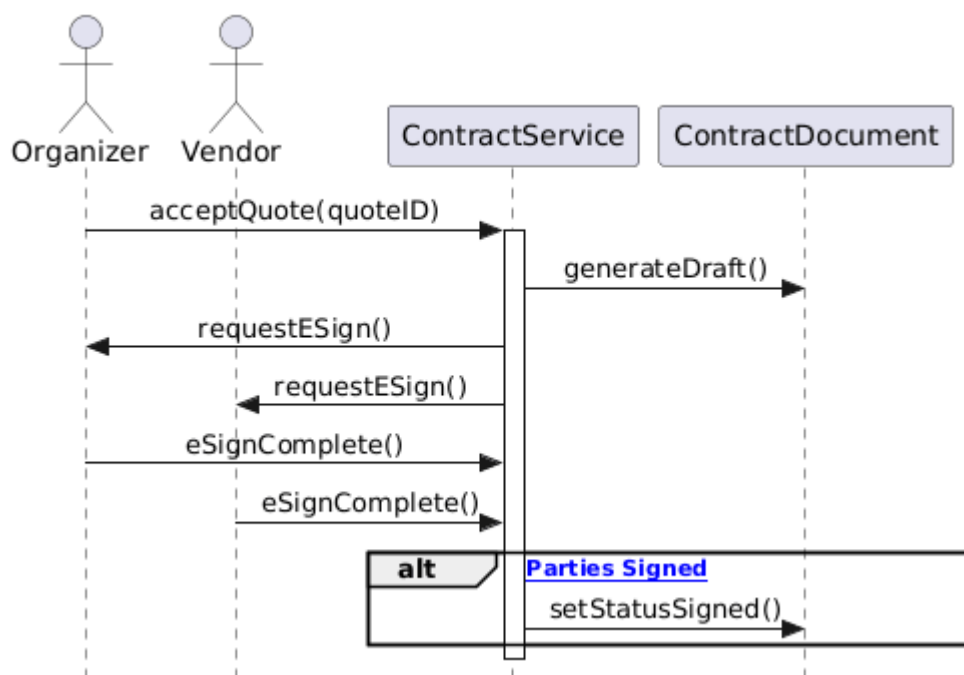


figure 12: Configure inventory & Enable sales

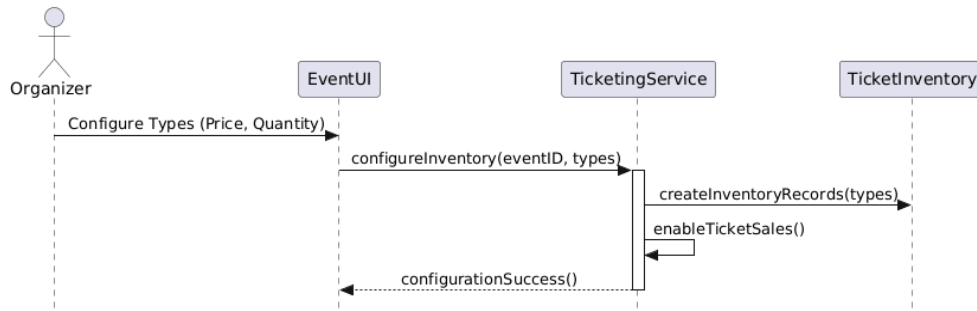


figure 13: Publish Ticket

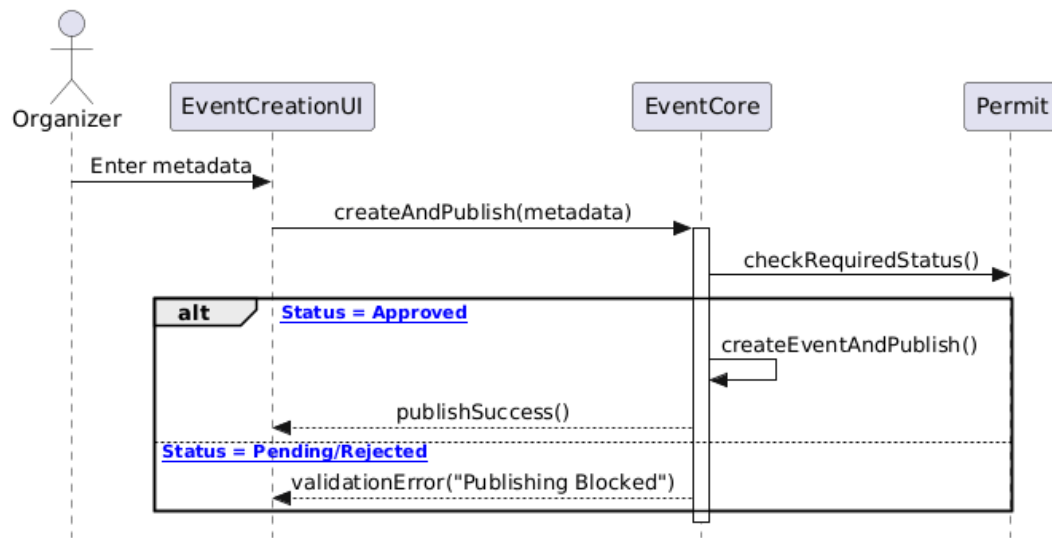


figure 14: Purchase Ticket

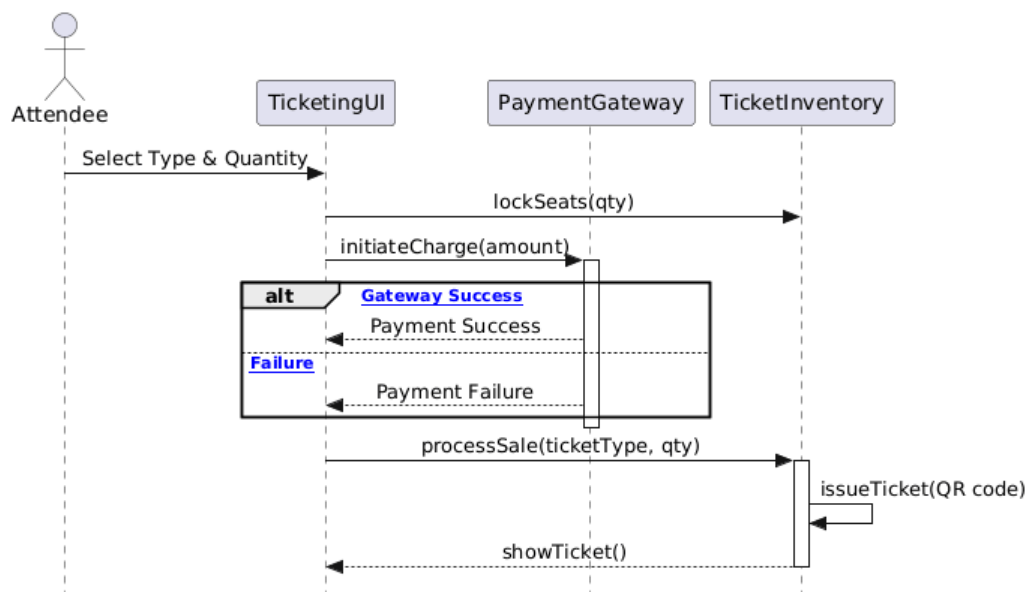


figure 15: Generate & Print Badge

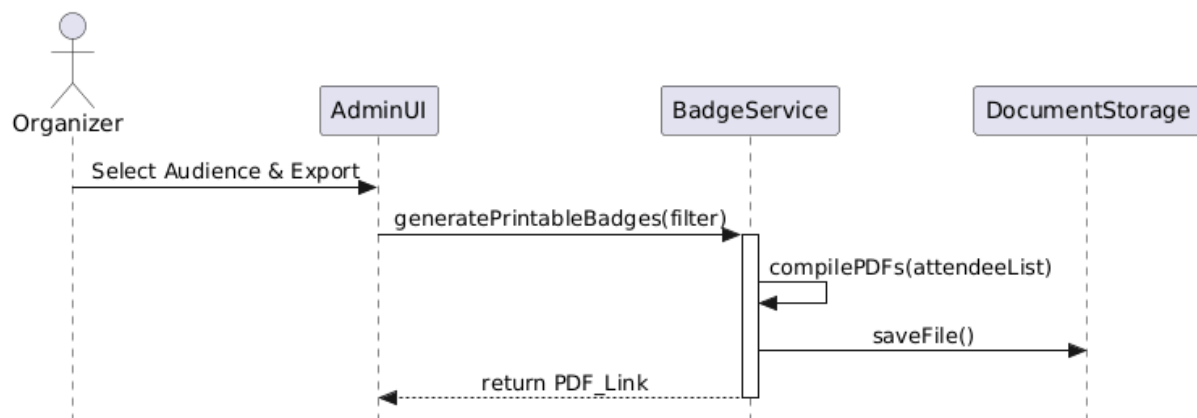


figure 16: On-site Incident Report

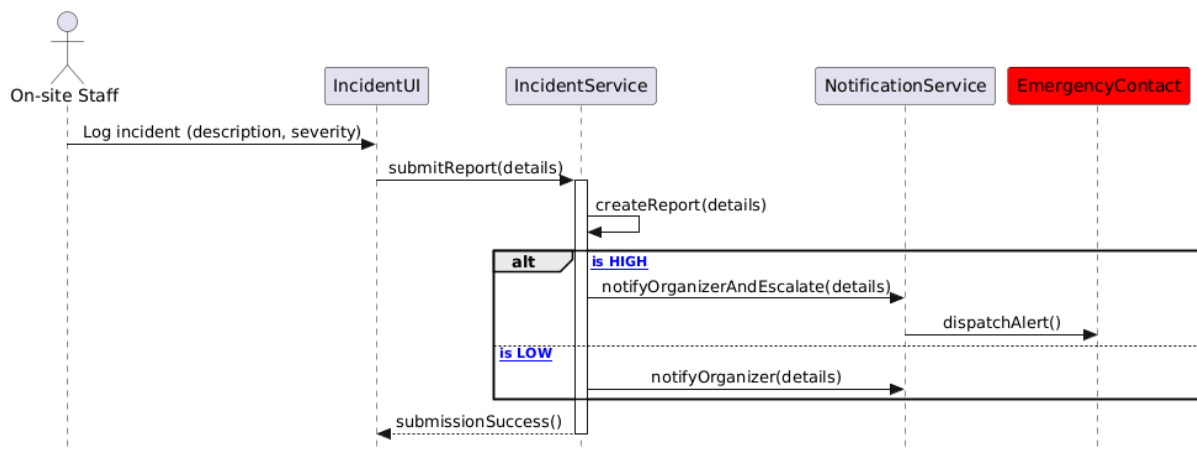


figure 17: Send Post-Event Survey

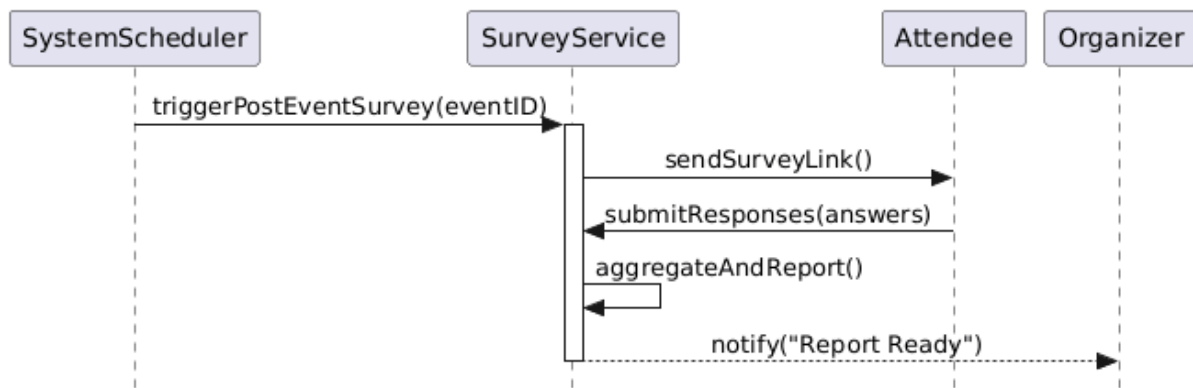
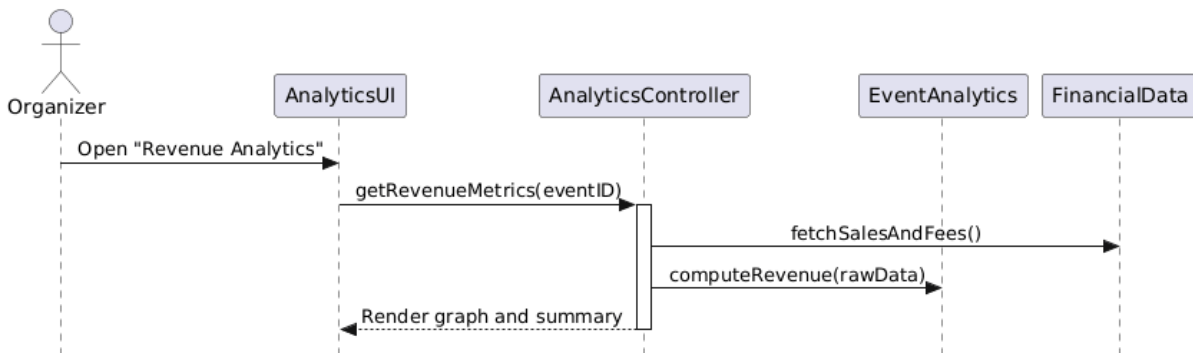


figure 18: Track Revenue Analytics



3.5.5 Activity Diagram

figure 19: Event Proposal and Ministry Verification (Licensing)

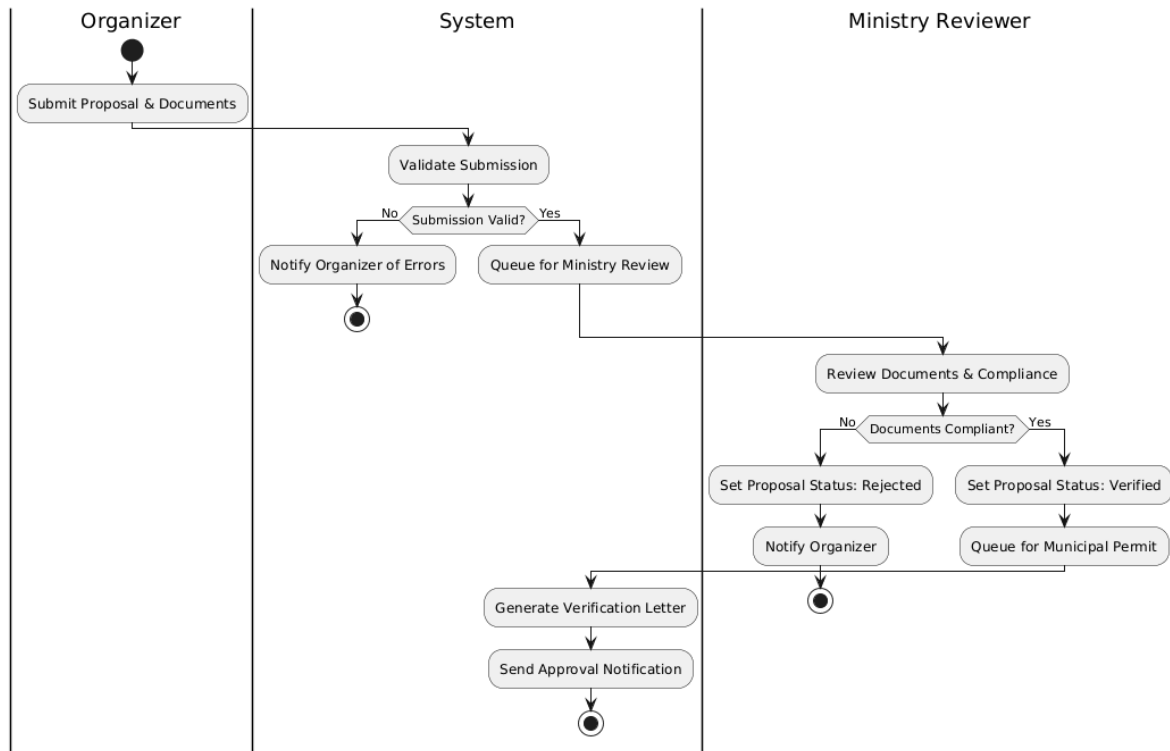


figure 20: Municipal Permit and Event Publishing

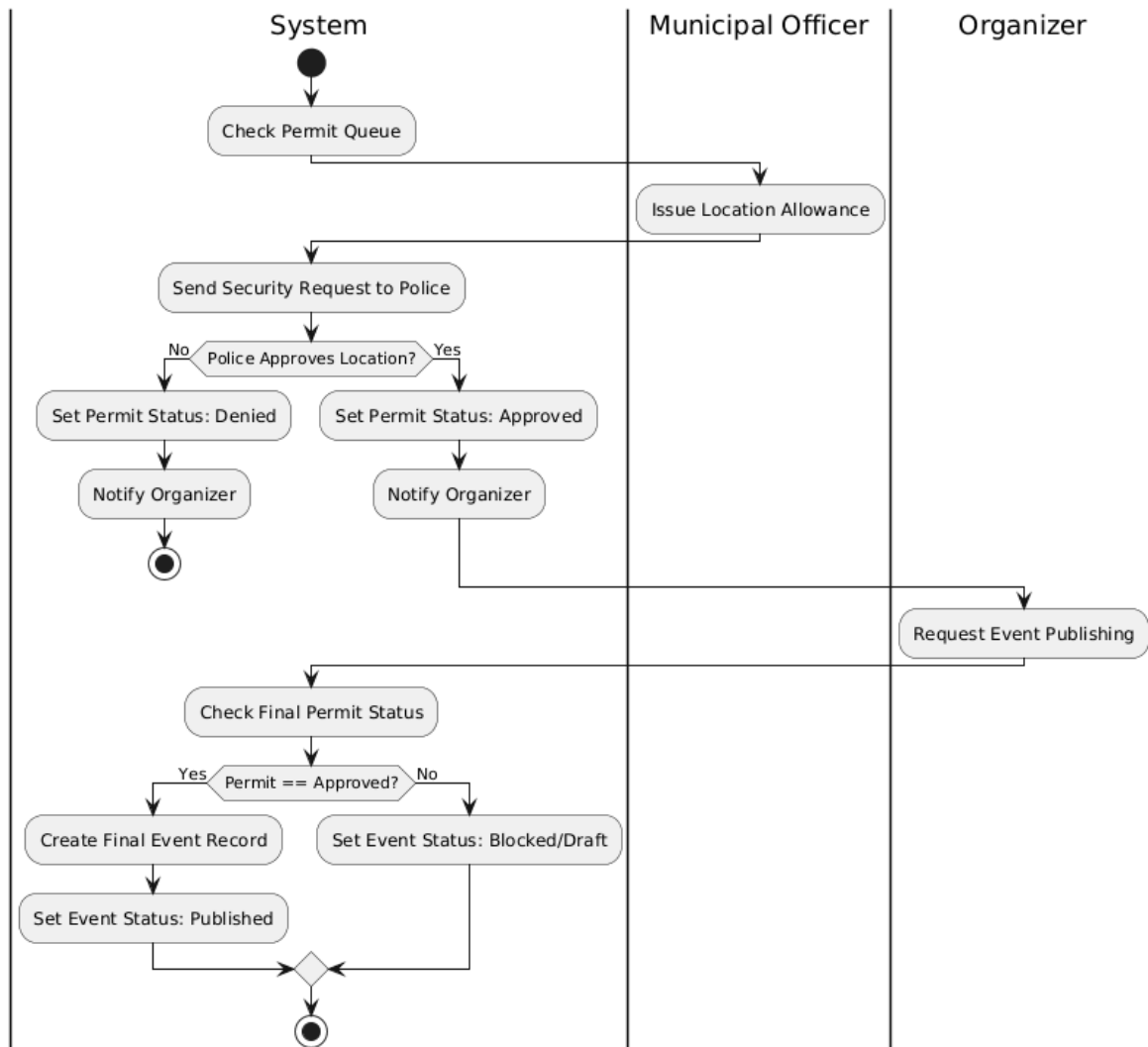


figure 21: AI-Assisted schedule generation and finalization

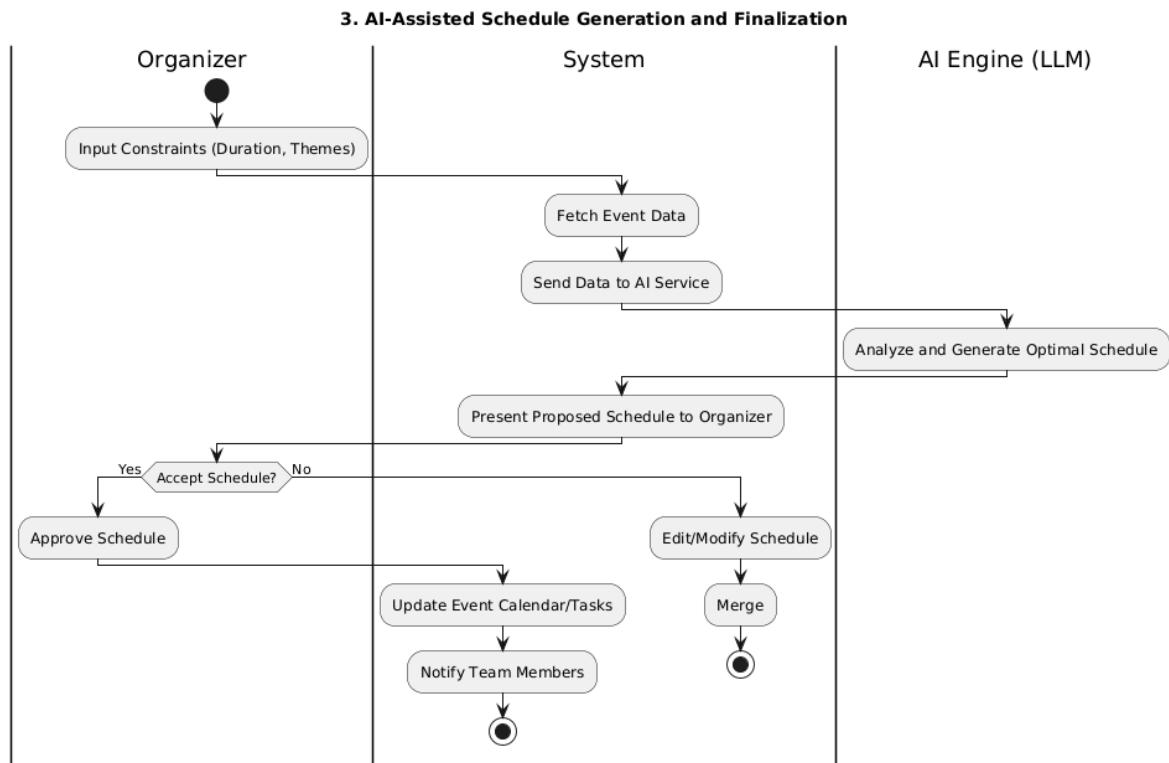


figure 22: Invitation and task assignment

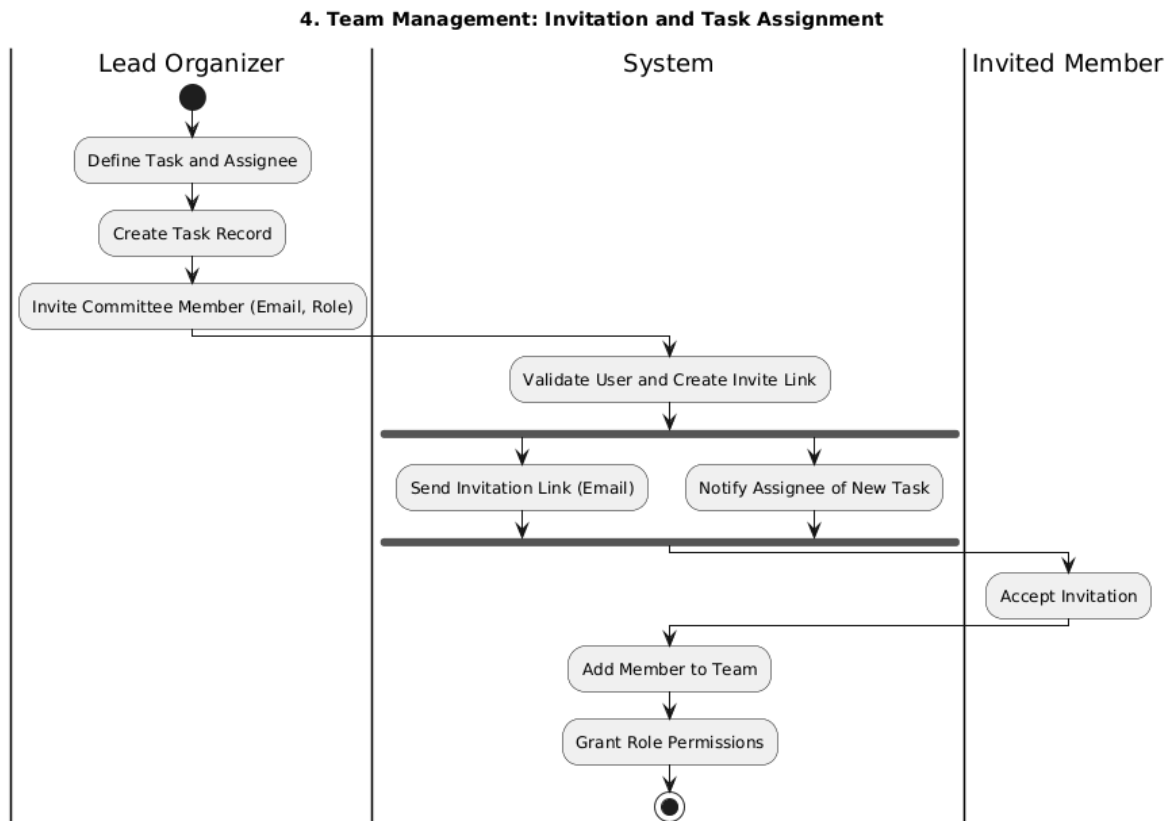


figure 23: Vendor registration and approval

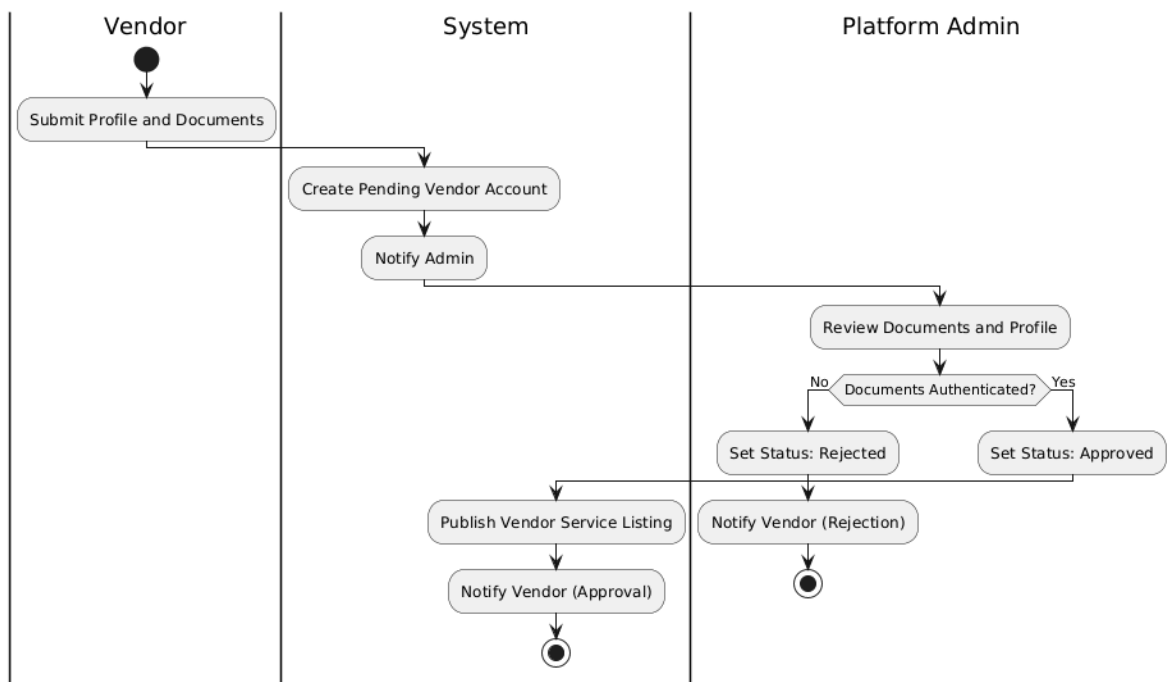


figure 24: Vendor quote acceptance and contract signing

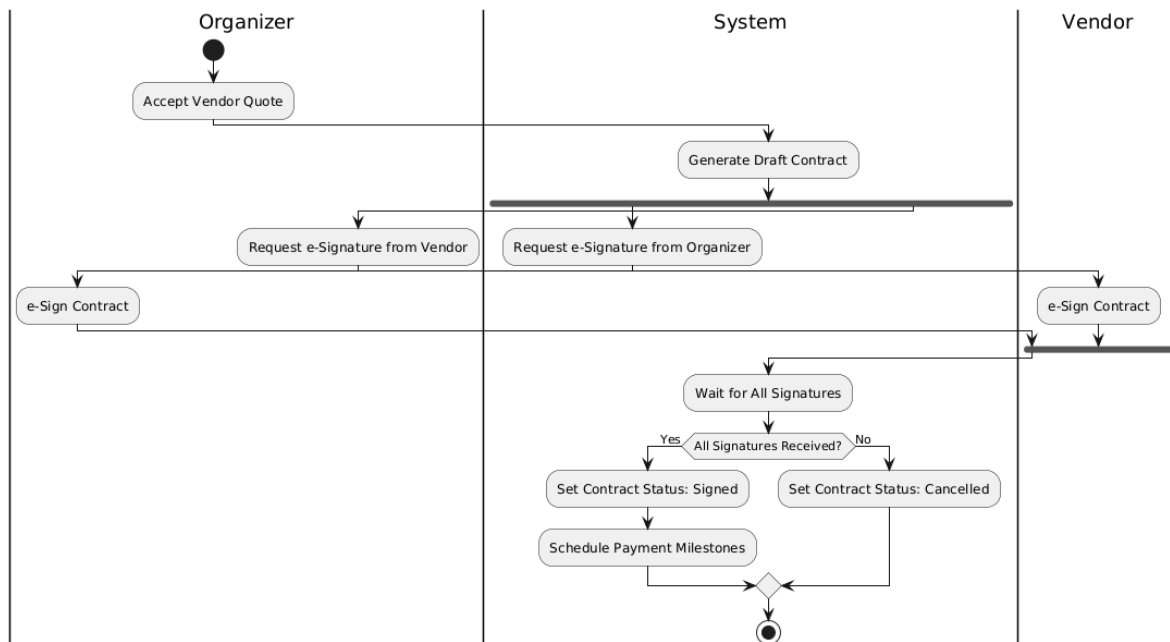


figure 25: Ticket purchase and inventory management

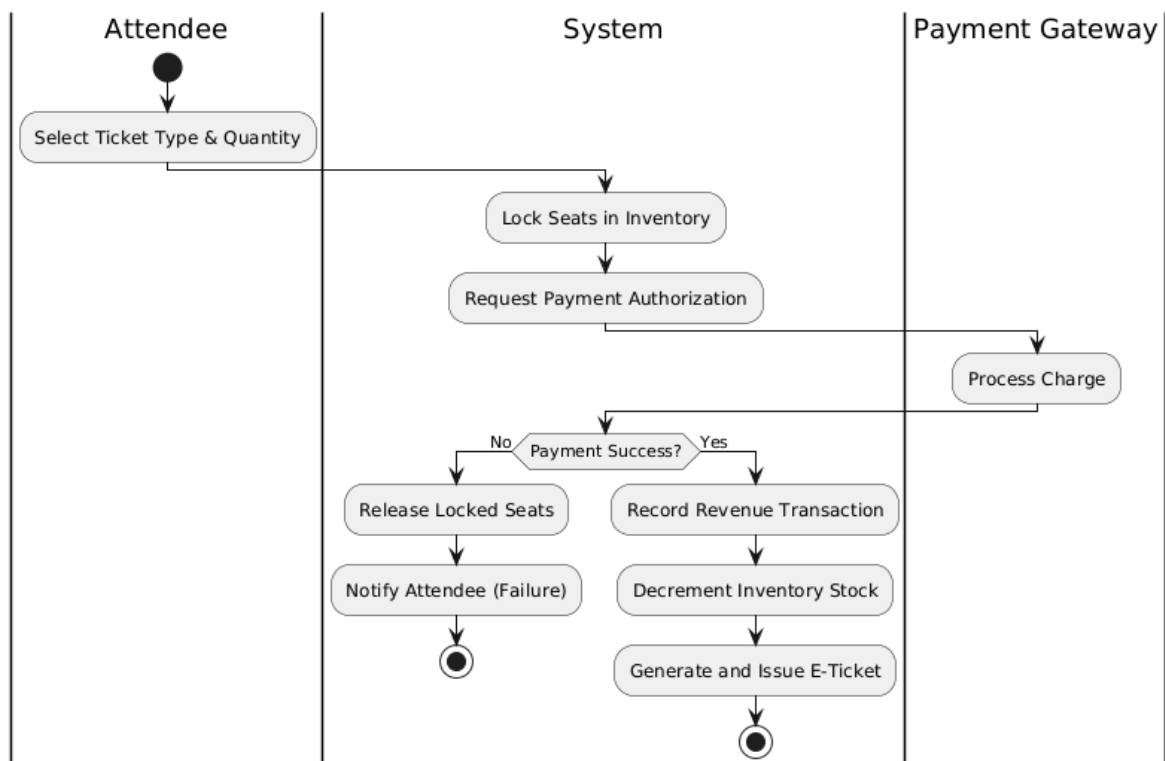


figure 26: On-site Attendee Check-in and Badge Issuance

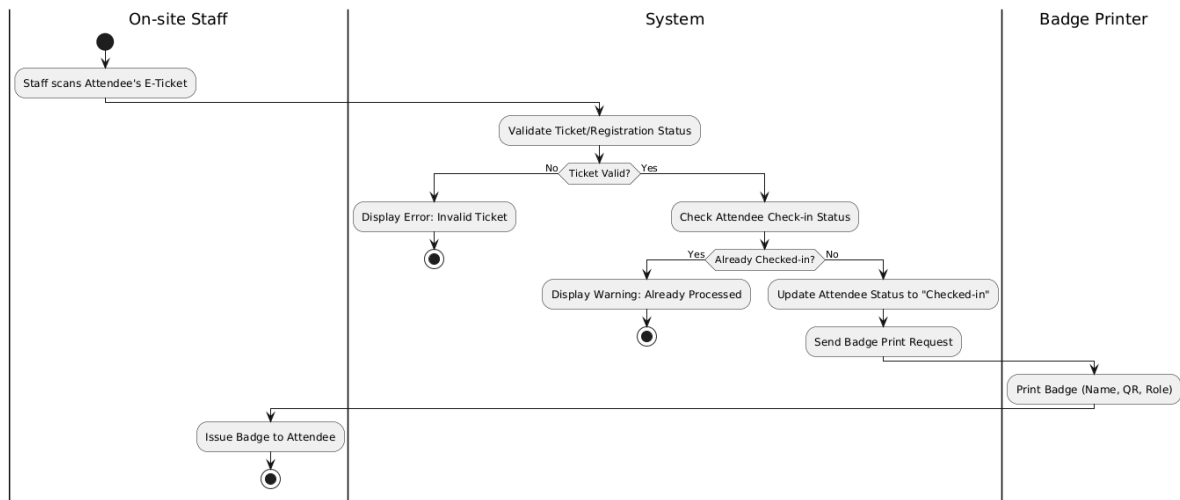
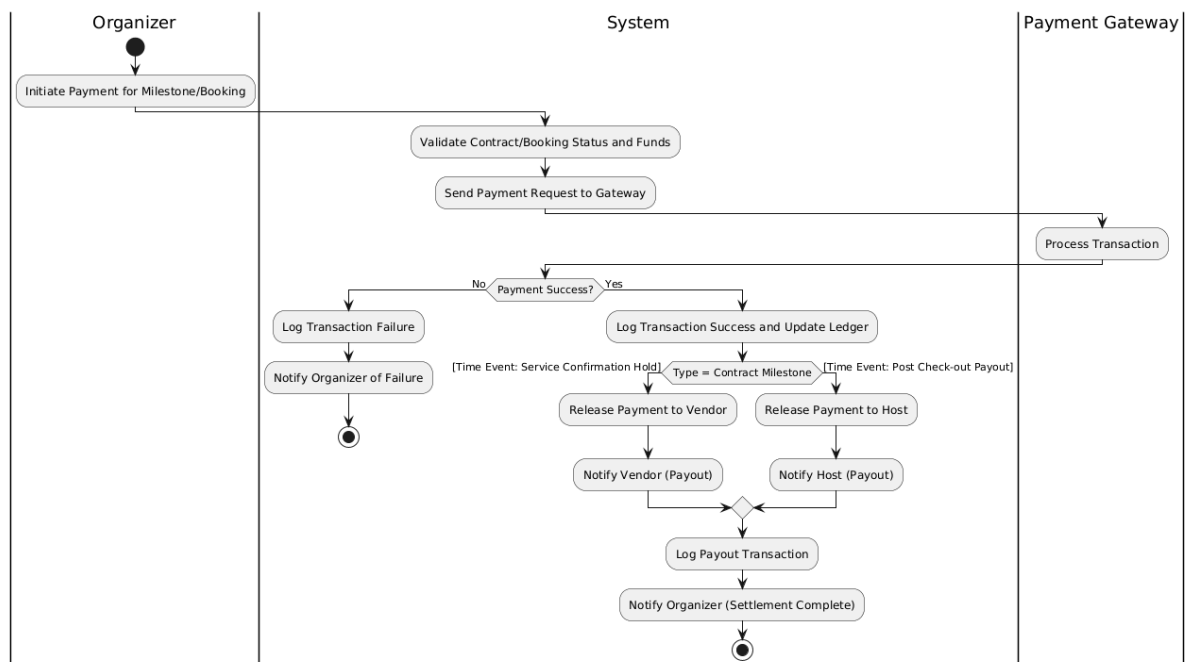


figure 27: Financial Settlement and Vendor Payout



3.5.6 State-Chart Diagram

figure 28: Event Proposal

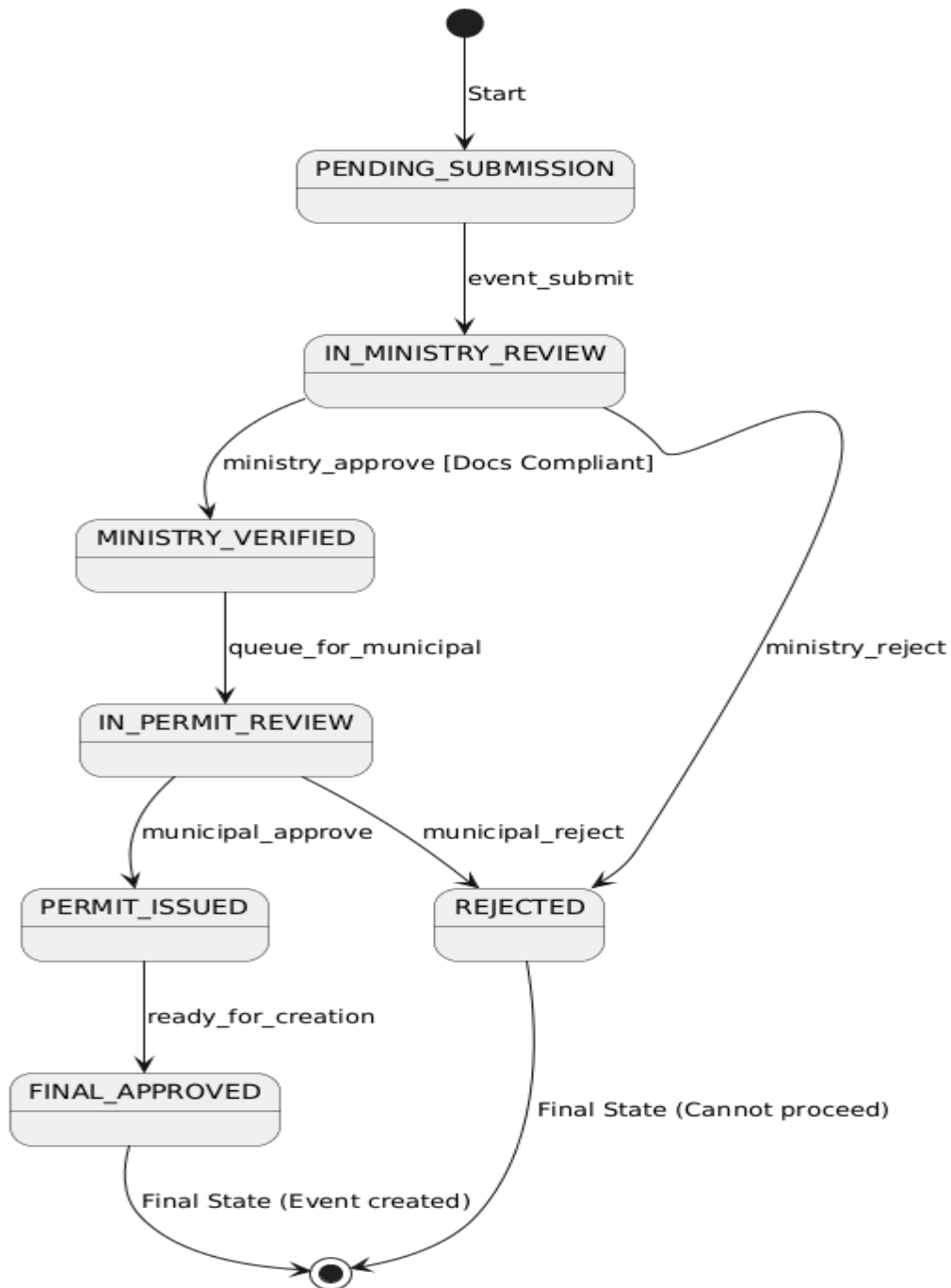


figure 29: Contract Engagement

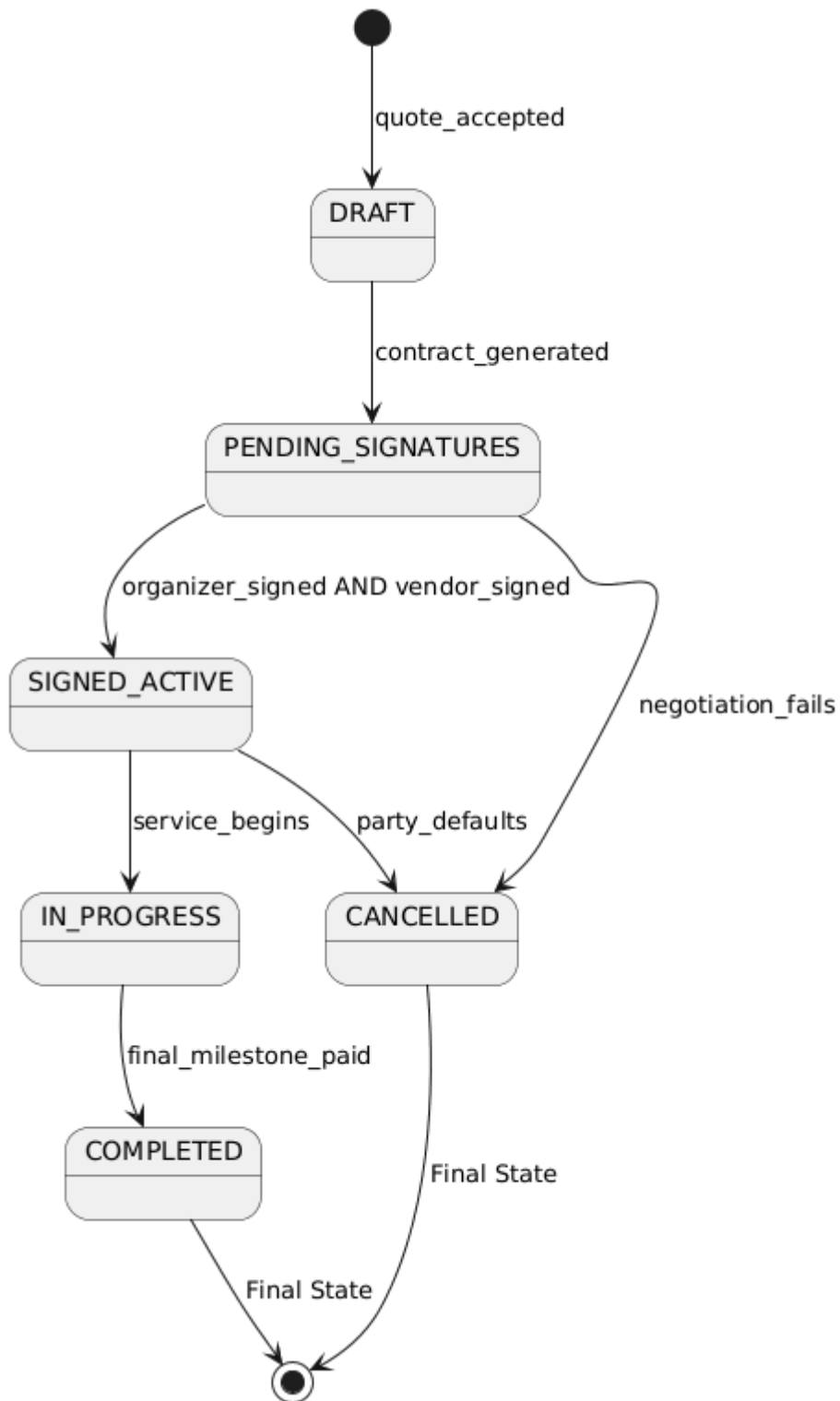


figure 30: Vendor Account Lifecycle

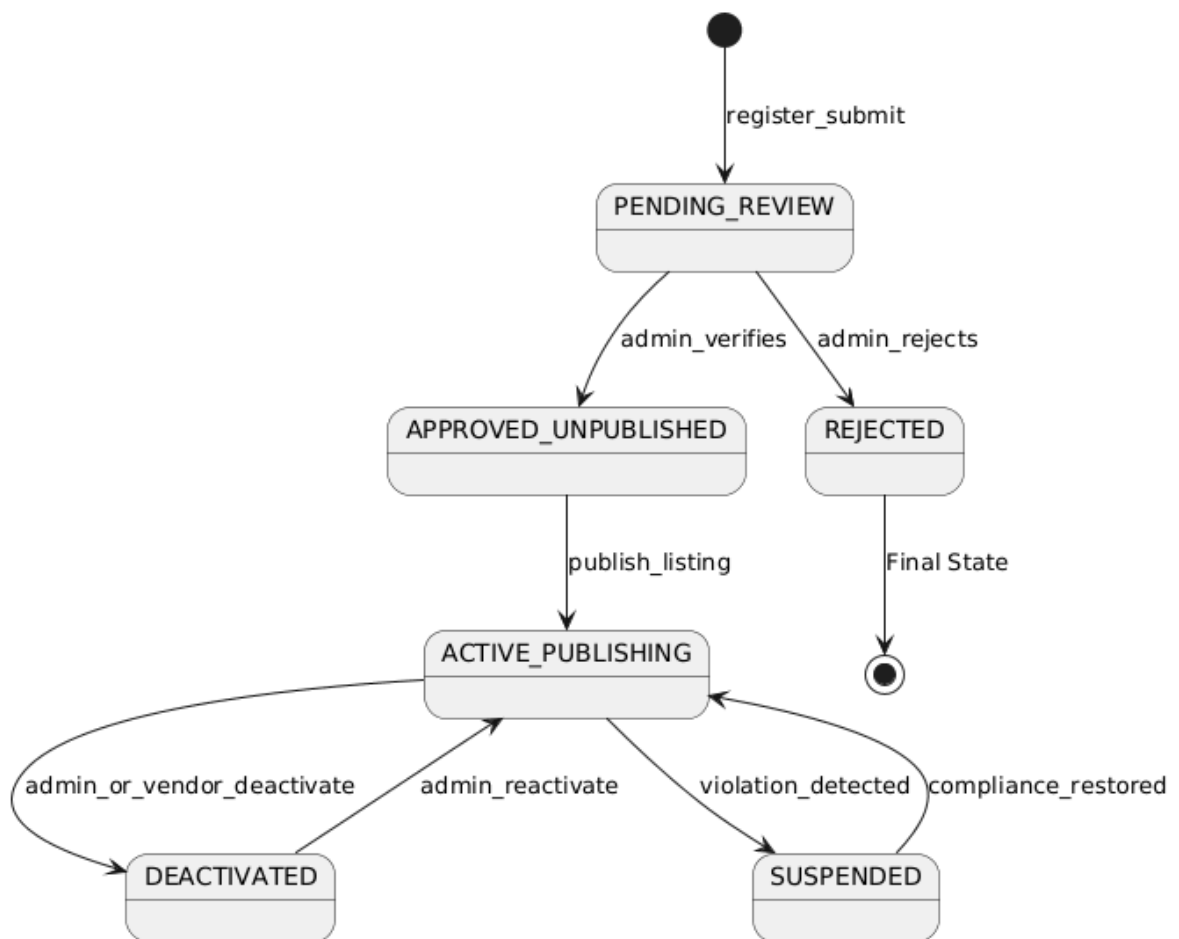


figure 31: Accommodation Booking

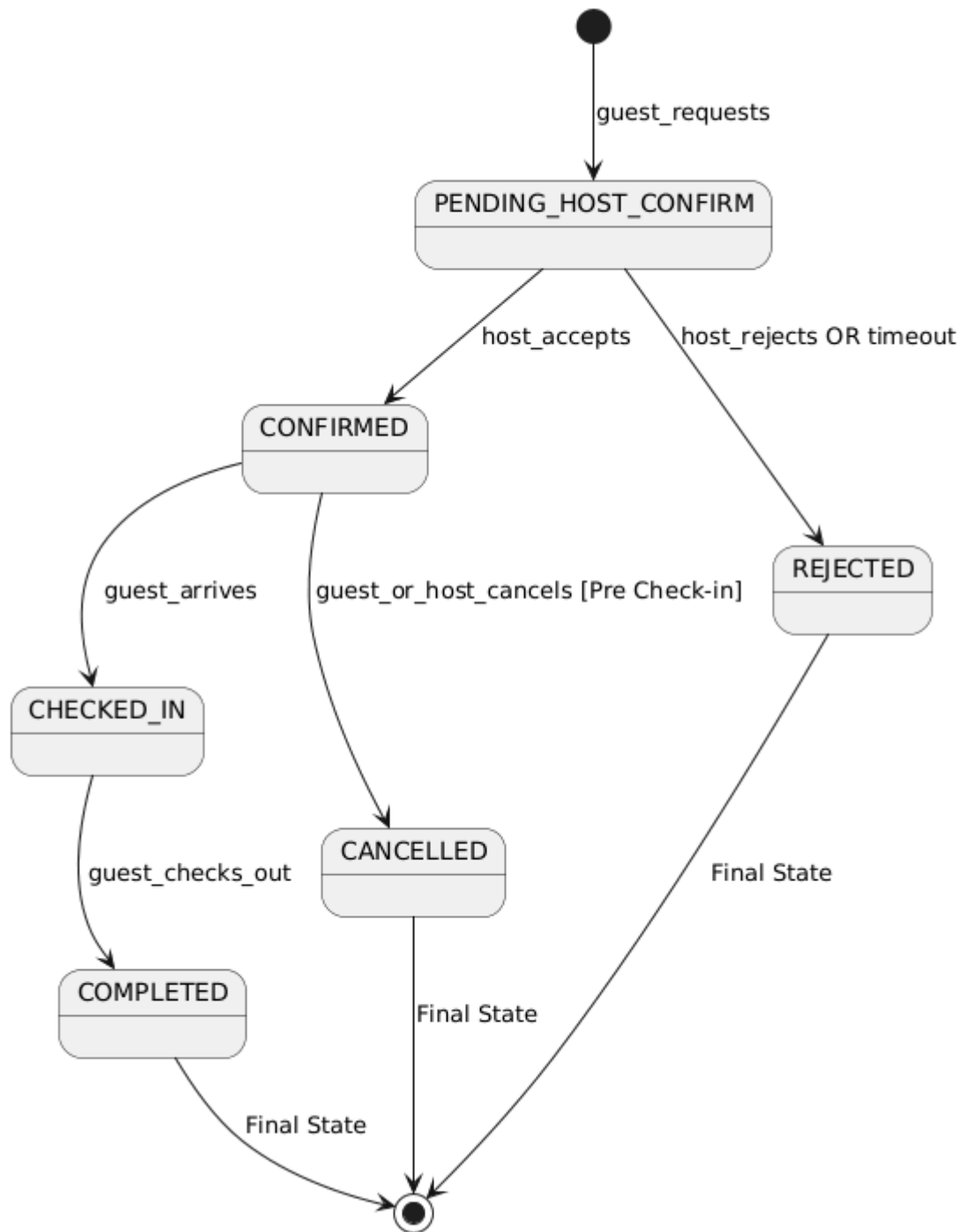
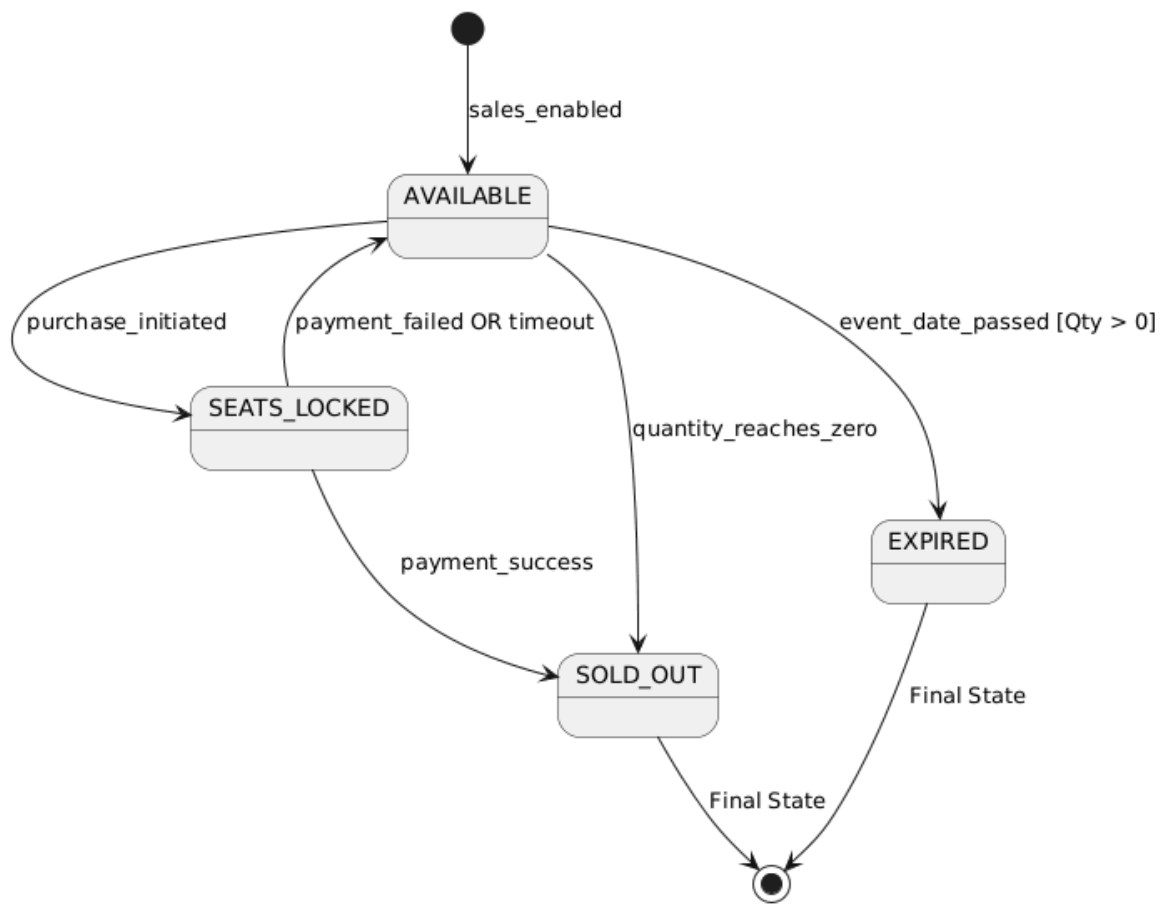


figure 32: Ticket Inventory Block



CHAPTER 4 : SYSTEM DESIGN

4.1. Overview

The system design of Global Connect Ethiopia International Professional Event Hub provides a structured blueprint that defines how the proposed platform operates, how its components interact, and how it fulfills both functional and non-functional requirements identified in earlier chapters. The design aims to translate conceptual requirements into an implementable architectural structure by defining the major modules, interfaces, data flows, and operational dependencies supporting end-to-end event management.

4.2. Purpose of the System Design

Specification (SRS) and system model, ensuring a structured, feasible, and well-aligned transition from conceptual analysis to implementation within the Ethiopian event ecosystem. The design decomposes core functionalities such as event proposal submission, government approval workflows, vendor registration and booking, ticketing and payments, scheduling, and on-site management into coherent subsystems that support consistent behavior across user roles, while also addressing critical non-functional requirements including performance, scalability, security, usability, availability, and data integrity through appropriate architectural patterns, database structures, and standardized interfaces. In addition, the system design aligns use cases and UML models with the technical architecture by mapping documented scenarios to implementation modules and defining how key entities are structured and managed within the database, thereby serving as a practical guide for development and testing, reducing ambiguity, improving efficiency, and supporting a maintainable and reliable codebase.

4.3. Design Goals

The design of Global Connect Ethiopia is guided by a set of goals and principles that ensure the system is robust, maintainable, scalable, and secure. These principles align with modern software engineering standards and support long-term sustainability of the platform.

1. Modularity

The system is divided into independent modules such as event management, licensing workflow, vendor marketplace, payment and ticketing, notifications, and analytics. Each module is responsible for a specific set of related features, making the system easier to maintain, test, and extend as new requirements emerge.

2. High Cohesion and Low Coupling

Subsystems are designed to perform closely related functions internally (high cohesion) while relying on well-defined API contracts to communicate with other modules (low coupling). This reduces dependency-related errors and allows updates to one component without disrupting others.

3. Abstraction and Encapsulation

Internal logic details such as database operations, validation rules, and pipeline processes are hidden behind service layers. Users and external modules interact only with exposed interfaces, improving security and reducing the risk of unintended modification.

4. Scalability and Performance

The system is designed to handle traffic spikes during major events, especially in ticket sales. Efficient data structures, optimized queries, asynchronous APIs, and caching mechanisms ensure that core pages load quickly and that ticketing operations maintain real-time accuracy without overbooking.

5. Security and Access Control

Security is a central design goal, addressed through encrypted communication (HTTPS), secure authentication mechanisms (JWT), hashed passwords, and strict role-based access control (RBAC). Government-related operations and VIP data are handled with additional safeguards to protect sensitive information.

6. Reusability

The platform uses reusable components UI templates, API utilities, database models, and validation schemas to minimize duplication, reduce bugs, and accelerate development. This also allows future teams to expand platform capabilities without rewriting existing logic.

7. Fault Tolerance and Reliability

The system design ensures proper error handling, data backup routines, and transaction safety. Ticketing and payment workflows employ transactional strategies to prevent duplicate payments or overbooking. Logs and failure alerts enable quick recovery from unexpected issues.

8. User-Centered Design

The platform is designed to be intuitive and accessible, supporting both English and Amharic users. Simple workflows, clear instructions, and responsive layouts ensure that organizers, vendors, and attendees can complete tasks with minimal training or technical knowledge.

Together, these design goals ensure that Global Connect Ethiopia provides a reliable, extensible, and efficient digital ecosystem that can grow with Ethiopia's evolving event industry while delivering a seamless experience for all stakeholders.

4.4 Proposed System Architecture

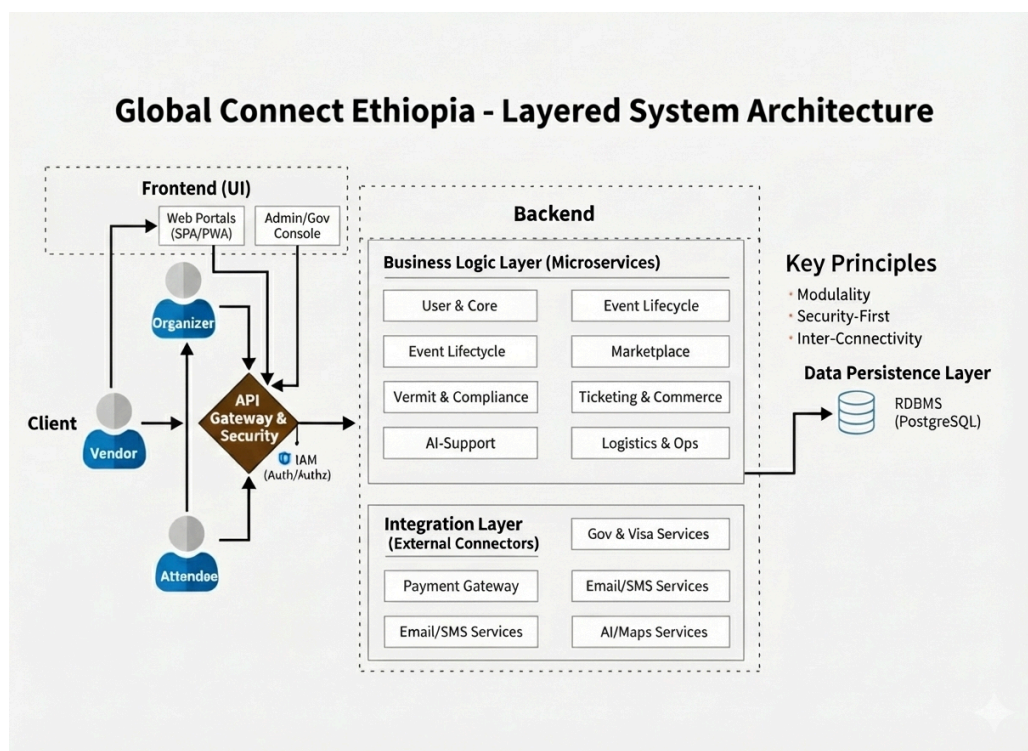
The proposed system architecture for Global Connect Ethiopia – International Professional Event Hub adopts a three-tier layered architecture. This architectural style separates the system into presentation, application, and data layers, ensuring clear separation of concerns, improved maintainability, scalability, and security.

The presentation layer provides a web-based user interface through which organizers, vendors, attendees, and government officials interact with the system. It is responsible for capturing user input, rendering system responses, and initiating requests to backend services through secure RESTful APIs. This layer contains no business logic, which helps keep the user interface lightweight and easy to evolve independently.

The application layer forms the core of the system and encapsulates all business logic and workflow coordination. It manages critical functionalities such as event creation and management, government licensing and approval processes, vendor coordination, ticketing,

payments, notifications, and role-based access control. The application layer also acts as a controlled gateway to external services such as payment gateways, notification services (email and SMS), and AI/LLM-based chatbot services. These external components are not modeled as separate architectural layers but are integrated through secure APIs, ensuring loose coupling, fault tolerance, and overall system stability.

The data layer is responsible for persistent storage and retrieval of system data. It stores user accounts, event details, licensing records, vendor information, ticketing data, and transaction records in a centralized database. All data access is mediated through the application layer, preventing direct client-side access and ensuring data integrity, security, and consistency.



4.5 System Decomposition

The system is organized into several high-level, domain-oriented subsystems to ensure modularity, scalability, and maintainability. These subsystems reflect the major functional and domain responsibilities of the system:

1. **User Management Subsystem**
2. **Event Management Subsystem**
3. **Approval and Governance Subsystem**

- 4. Vendor and Service Management Subsystem**
- 5. Ticketing Subsystem**
- 6. Payment Subsystem**
- 7. Resource and Accommodation Subsystem**
- 8. Feedback and Reporting Subsystem**

4.6 Subsystem Description

4.6.1 User Management Subsystem

The User Management Subsystem provides the foundational identity and access layer for the Global Connect Ethiopia platform. It is responsible for managing all system users, including event organizers, vendors, attendees, government officials, and administrators. The subsystem handles user registration, secure authentication, profile management, account verification, and session lifecycle control. By maintaining a centralized user repository, the system ensures that every interaction within the platform is traceable to an authenticated and authorized user.

In addition to identity management, this subsystem enforces role-based access control (RBAC) across the system. Each user is assigned a predefined role that determines their permitted actions and accessible resources. Security mechanisms such as password hashing, token-based authentication, and multi-factor authentication for sensitive roles are applied to protect user data and prevent unauthorized access. As a result, the User Management Subsystem acts as a security gatekeeper and enables personalized system behavior based on user roles.

Interaction with Other Subsystems

- Provides authentication and authorization services to all subsystems
- Supplies user identity and role information to the Event Management Subsystem
- Validates government officials for the Approval and Governance Subsystem
- Links vendor accounts in the Vendor and Service Management Subsystem
- Identifies buyers and participants in the Ticketing and Payment Subsystems

4.6.2 Event Management Subsystem

The Event Management Subsystem is the central operational core of the platform and manages the complete lifecycle of events. It enables organizers to create event proposals, define event details, manage schedules, publish approved events, and monitor event progress until completion. All event-related data, including status, capacity, schedules, and descriptions, are maintained within this subsystem, making it the authoritative source of event information.

This subsystem coordinates complex workflows that span multiple phases, such as proposal submission, regulatory approval, vendor engagement, ticket sales, and post-event analysis. By maintaining consistent event state transitions, the subsystem ensures that actions such as ticket sales or vendor contracts can only occur when the event satisfies predefined conditions, such as regulatory approval or publication status.

Interaction with Other Subsystems

- Sends proposals and receives decisions from the Approval and Governance Subsystem
- Associates vendors and services via the Vendor and Service Management Subsystem
- Supplies event context to the Ticketing Subsystem
- Provides event metadata to the Payment Subsystem
- Links venues and accommodations through the Resource and Accommodation Subsystem
- Feeds event data into the Feedback and Reporting Subsystem

4.6.3 Approval and Governance Subsystem

The Approval and Governance Subsystem supports regulatory compliance by digitizing the event licensing and approval process. It allows authorized government and municipal officials to review submitted event proposals, verify compliance with legal, safety, and administrative regulations, and issue approval or rejection decisions. This subsystem ensures that all regulatory actions are formally recorded and traceable.

By automating approval workflows, the subsystem reduces manual paperwork, improves transparency, and enforces standardized decision-making processes. Approved events are

issued official digital permits, while rejected or flagged proposals are returned with documented feedback. This structured governance mechanism strengthens trust between event organizers and regulatory authorities.

Interaction with Other Subsystems

- Retrieves proposals from the Event Management Subsystem
- Verifies official roles through the User Management Subsystem
- Updates event approval status in the Event Management Subsystem
- Triggers system notifications through shared infrastructure services

4.6.4 Vendor and Service Management Subsystem

The Vendor and Service Management Subsystem manages all interactions between event organizers and service providers. It enables vendors to register on the platform, submit compliance documents, list service offerings, and respond to quotation requests. Organizers can browse vendors, compare services, request quotes, and establish contractual relationships digitally.

This subsystem formalizes vendor engagement by supporting structured workflows such as quote submission, acceptance, contract signing, and service tracking. By digitizing these processes, the platform ensures accountability, reduces coordination overhead, and improves service transparency for large and complex events.

Interaction with Other Subsystems

- Links vendor accounts through the User Management Subsystem
- Associates vendor services with events via the Event Management Subsystem
- Coordinates vendor payments with the Payment Subsystem
- Provides vendor performance data to the Feedback and Reporting Subsystem

4.6.5 Ticketing Subsystem

The Ticketing Subsystem is responsible for managing event attendance through structured ticket operations. It allows organizers to configure ticket types, pricing tiers, and sales rules, while enabling attendees to reserve and purchase tickets digitally. The subsystem maintains

real-time inventory tracking to prevent overbooking and ensures that ticket availability reflects actual capacity.

In addition, the subsystem generates digital tickets with QR codes and supports on-site check-in and badge issuance. These features improve operational efficiency during events and provide accurate attendance tracking, which is critical for safety, reporting, and post-event analysis.

Interaction with Other Subsystems

- Retrieves event and capacity data from the Event Management Subsystem
- Confirms payment completion through the Payment Subsystem
- Identifies attendees via the User Management Subsystem
- Supplies attendance data to the Feedback and Reporting Subsystem

4.6.6 Payment Subsystem

The Payment Subsystem manages all financial transactions within the platform. It securely processes payments related to ticket purchases, accommodation bookings, and vendor settlements through integrated payment gateways. Each transaction is recorded with detailed metadata to ensure traceability, transparency, and auditability.

By decoupling financial logic from operational workflows, the subsystem enhances security and scalability. It also supports payment validation, transaction status tracking, and refund handling, ensuring consistent financial operations across the platform.

Interaction with Other Subsystems

- Confirms ticket purchases for the Ticketing Subsystem
- Processes vendor settlements for the Vendor and Service Management Subsystem
- Handles accommodation payments for the Resource and Accommodation Subsystem
- Provides financial data to the Feedback and Reporting Subsystem

4.6.7 Resource and Accommodation Subsystem

The Resource and Accommodation Subsystem manages physical and logistical resources required for events, including venues, hotels, and temporary accommodations. Resource

providers can publish availability, while organizers and attendees can browse options and make reservations aligned with event schedules.

This subsystem ensures efficient allocation of resources by preventing scheduling conflicts and maintaining booking consistency. By integrating accommodation management into the event lifecycle, the platform enhances participant experience and reduces external coordination complexity.

Interaction with Other Subsystems

- Links resources to events via the Event Management Subsystem
- Identifies hosts and guests through the User Management Subsystem
- Processes payments using the Payment Subsystem
- Supplies booking data to the Feedback and Reporting Subsystem

4.6.8 Feedback and Reporting Subsystem

The Feedback and Reporting Subsystem supports post-event evaluation and operational insight generation. It collects attendee surveys, incident reports, and system metrics to assess event performance, user satisfaction, and financial outcomes.

The subsystem aggregates data across multiple subsystems to generate analytical reports for organizers and administrators. These insights support evidence-based decision-making, continuous system improvement, and strategic planning for future events.

Interaction with Other Subsystems

- Associates reports with events from the Event Management Subsystem
- Uses attendance data from the Ticketing Subsystem
- Integrates revenue data from the Payment Subsystem
- Evaluates vendor performance from the Vendor and Service Management Subsystem
- Links feedback to users via the User Management Subsystem

4.7 Persistent Data Management

The database architecture for the Global Connect Ethiopia system is built upon a robust NoSQL foundation, adhering to modern development practices and the dynamic needs of professional event management.

Data Storage Strategy

The system utilizes **MongoDB**, a document-oriented NoSQL database, chosen for its intrinsic ability to manage persistent data with flexible document structures, support efficient embedding of related entities, and provide horizontal scalability. This design prioritizes data locality, schema flexibility, and query efficiency, ensuring alignment with the rapidly changing data landscape of event management in Ethiopia. Data consistency and integrity are primarily enforced at the application layer, leveraging MongoDB's indexing capabilities and unique identifiers (`_id`) to ensure reliable operations.

Data Relationships

The Global Connect Ethiopia database organizes data through logical relationships between these collections, balancing performance, scalability, and flexibility:

User Centrality: The Users collection serves as the central hub, linked to every core process: event proposals, venues, vendors, orders, tickets, and team tasks.

Event Lifecycle: Event Proposals mature into approved Events. Events, in turn, form the backbone of the system, relating to Venues, Tickets, Accommodation Listings, and all transactional and compliance-focused collections.

Transactional Flow: The vendor process flows from Quotes to finalized Contracts. Orders document all purchases, linking user actions (purchases) directly to Tickets and Events.

Monitoring and Feedback: Surveys, Survey Responses, Incident Reports, and Team Tasks are all associated with Events to capture feedback, monitor activities, and generate audit logs.

Data Integrity and Optimization

The database implementation utilizes the following techniques to ensure data integrity and performance:

Unique Identifiers: Every document is secured with a primary `_id` field, typically an `ObjectId`. References between collections strictly use this `_id` for logical consistency.

Indexing: Critical foreign key fields, such as `userId`, `eventId`, `organizerId`, and `proposalId`, are indexed to significantly boost read performance across frequently queried paths.

Embedded Documents: Embedding is strategically used to reduce join operations and improve read efficiency for entities that are tightly coupled (e.g., embedding critical ticket details within an `Order` document).

Scalability: The schema is designed to inherently support horizontal scaling of collections, accommodating the projected growth in users, events, and transactions without requiring complex restructuring.

This design methodology enables the platform to manage complex event workflows while maintaining a clear, auditable, and high-performance data architecture.

1. Users Collection

Attribute Name	Type / Pattern
<code>_id</code>	<code>ObjectId</code>
<code>email</code>	<code>String</code>
<code>password_hash</code>	<code>String</code>
<code>role</code>	<code>String</code>
<code>first_name, last_name</code>	<code>String</code>

phone_number	String
organization_name	String
is_verified	Boolean
profile_picture_url	String

2. Events Collection

Attribute Name	Type / Pattern
_id	ObjectId
organizer_id	ObjectId (Refers to Users)
title, description	String
category	String
location	String
start_date, end_date	Date
status	String
proposal_id	ObjectId (Refers to EventProposals)
image_urls	Array of String
ticket_types	Array of Object

3. EventSchedules Collection

Attribute Name	Type / Pattern
<code>_id</code>	ObjectId
<code>event_id</code>	ObjectId (Refers to Events)
<code>session_title, description</code>	String
<code>start_time, end_time</code>	Date
<code>speaker_id</code>	ObjectId (Refers to Users)
<code>room_location</code>	String
<code>is_ai_suggestion</code>	Boolean

4. TicketPurchases Collection

Attribute Name	Type / Pattern
<code>_id</code>	ObjectId
<code>event_id</code>	ObjectId (Refers to Events)
<code>attendee_id</code>	ObjectId (Refers to Users)
<code>ticket_type_name</code>	String

quantity	Number
total_amount	Decimal
payment_id	ObjectId (Refers to Payments)
check_in_status	String
purchase_date	Date

5. Vendors Collection

Attribute Name	Type / Pattern
_id	ObjectId
user_id	ObjectId (Refers to Users)
company_name	String
service_category	String
description	String
registration_status	String
average_rating	Decimal
license_document_url	String

6. VendorServices Collection

Attribute Name	Type / Pattern
_id	ObjectId
vendor_id	ObjectId (Refers to Vendors)
name, description	String
price	Decimal
unit_of_measure	String
images	Array of String

7. EventProposals Collection

Attribute Name	Type / Pattern
_id	ObjectId
event_id	ObjectId (Refers to Events)
organizer_id	ObjectId (Refers to Users)
submission_date	Date
current_stage	String

documents	Array of Object
review_history	Array of Object
final_status	String

8. Payments Collection

Attribute Name	Type / Pattern
_id	ObjectId
transaction_ref	String
source_collection(origin of the payment)	String
source_id	ObjectId
user_id (Participant or Organizer)	ObjectId (Refers to Users)
amount, currency	Decimal, String
status	String
timestamp	Date

9. Contracts Collection

Attribute Name	Type / Pattern
_id	ObjectId
event_id	ObjectId (Refers to Events)
vendor_id	ObjectId (Refers to Vendors)
organizer_id	ObjectId (Refers to Users)
contract_details	Object
agreed_price	Decimal
status	String
document_url	String

10. Reviews Collection

Attribute Name	Type / Pattern
_id	ObjectId
vendor_id	ObjectId (Refers to Vendors)
reviewer_id	ObjectId (Refers to Users)
rating	Number
comment	String

review_date	Date
is_approved	Boolean

4.8. Component Diagram

4.8.1 Components

The Global Connect Ethiopia platform is built upon a layered architectural model, decomposing the system into four primary components that manage the presentation, business logic, persistence, and specialized intelligent services.

1. Frontend (Client)

- Technology: React.js-based User Interface.
- Responsibility: Responsible for presenting data and handling user interactions.
- Modules include:
 - User Interface: Dashboards, forms, and navigation.
 - Event Proposal UI: Submission forms, document uploads, and status tracking.
 - Marketplace UI: Vendor browsing, quote requests, and contract signing.
 - Ticketing UI: Ticket type configuration, purchase checkout, and inventory views.
 - Operations UI: Accommodation listings, badge generation, incident reporting, and survey distribution.
 - Payment UI: Secure checkout pages, payment method selection, transaction confirmation, and payment status feedback.
- Communication: Communicates with the backend through REST APIs for all operations.

2. Backend (Server)

- Technology: FastAPI with Python.
- Responsibility: Manages business logic, handles user requests, and interacts with both the database and external AI services.
- Modules include:
 - Authentication & Authorization: Verifies user credentials and manages roles (e.g., Organizer, Vendor, Attendee).
 - Proposal Management API: Handles creation, review, and approval workflows for event proposals.

- Event Management API: Supports publishing, scheduling, and venue reservations.
- Marketplace API: Processes quotes, negotiations, and contract e-signing.
- Ticketing & Payment API: Manages ticket orders, inventory locking, and payment integrations.
- Operations API: Facilitates accommodation bookings, badge issuance, incident logging, and analytics aggregation.
- Ticketing API: handles ticket configuration, pricing, inventory management, ticket reservations, digital ticket generation, and attendee check-in. It ensures accurate attendance tracking and prevents overbooking.
- Payment API: processes all financial transactions within the system, including ticket purchases, accommodation payments, vendor settlements, and refunds. It integrates with external payment gateways and maintains transaction records.

3. Database (MongoDB)

- Technology: NoSQL Database.
- Responsibility: Manages persistent storage of user profiles, event data, proposals, tickets, contracts, and operational records.
- Integrity: Enforces data consistency and integrity through structured schemas.

4. External AI Services (e.g., Grok API)

- Technology: AI Providers (e.g., xAI Grok models).
- Role: The backend makes API calls to external AI services to offload tasks such as:
 - Schedule generation based on event constraints.
 - Regulatory inquiries and compliance recommendations.
- Process: These AI services process requests and return results in a structured format (typically JSON). The backend then processes this data and delivers it to the frontend, ensuring it is ready for user interaction.
- Secure Communication:
 - All interactions with external AI services are encrypted to ensure data privacy and maintain compliance with relevant security standards.
 - This encryption ensures that sensitive user data is protected during transmission between the backend and external AI providers.

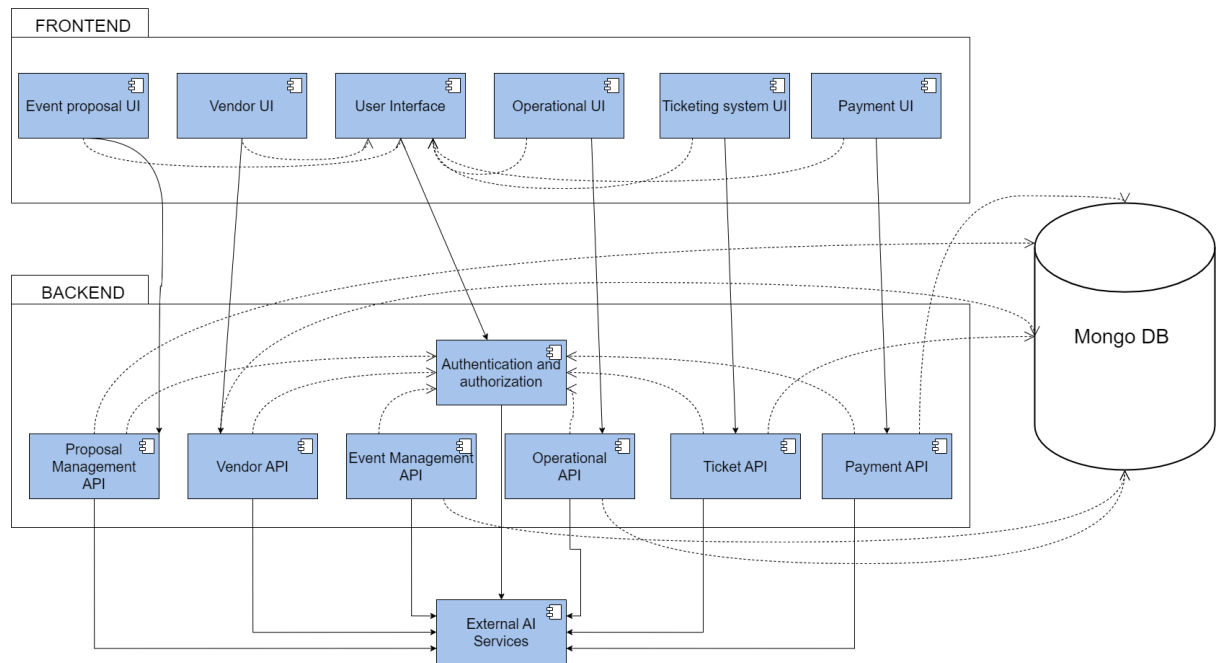


figure 36. Component diagram

4.9. Database Diagram

This section presents the database structure of the proposed system, outlining its tables, columns, keys, and relationships. The database design uses a relational model to ensure data integrity, scalability, and efficient access. Below is a detailed description of each table in the database.

4.9.1. Entities and Attributes

Table 10: Entities and their attributes

Entity	Attributes
User	user_id (PK), full_name, email, phone, password_hash, role, status, bio, organization, created_at

EventProposal	proposal_id (PK), organizer_id (FK), title, event_type, budget_estimate, expected_attendees, start_date, end_date, status
ProposalReview	review_id (PK), proposal_id (FK), reviewer_id (FK), decision, comments, reviewed_at
Permit	permit_id (PK), proposal_id (FK), city_status, police_status, issued_at
Document	document_id (PK), owner_id (FK), file_url, document_type, uploaded_at
Event	event_id (PK), proposal_id (FK), organizer_id (FK), venue_id (FK), title, description, capacity, status, schedule_json
Venue	venue_id (PK), provider_id (FK), name, capacity, location, availability
Reservation	reservation_id (PK), event_id (FK), venue_id (FK), status, reserved_at
Vendor	vendor_id (PK), user_id (FK), business_name, tin, national_id, status, rating
VendorService	service_id (PK), vendor_id (FK), title, description, price_range, availability
Quote	quote_id (PK), event_id (FK), vendor_id (FK), service_id (FK), amount, status, created_at

Contract	contract_id (PK), event_id (FK), vendor_id (FK), terms, signed_by_vendor, signed_by_organizer, payment_milestones, created_at
TicketType	ticket_type_id (PK), event_id (FK), name, price, quantity_total, quantity_available
Ticket	ticket_id (PK), ticket_type_id (FK), purchaser_id (FK), payment_id (FK), qr_code, status, issued_at
Payment	payment_id (PK), user_id (FK), amount, gateway, transaction_ref, status, created_at
AccommodationListing	listing_id (PK), host_id (FK), event_id (FK), title, price, capacity, amenities, status
AccommodationBooking	booking_id (PK), listing_id (FK), user_id (FK), start_date, end_date, status
Badge	badge_id (PK), event_id (FK), user_id (FK), role, qr_code, checked_in
IncidentReport	incident_id (PK), event_id (FK), reporter_id (FK), severity, description, reported_at
Survey	survey_id (PK), event_id (FK), title, questions
SurveyResponse	response_id (PK), survey_id (FK), user_id (FK), answers, submitted_at

TeamTask	task_id (PK), event_id (FK), assigned_to (FK), title, due_date, status
EventAnalytics	analytics_id (PK), event_id (FK), metrics, generated_at
AIInteraction	ai_id (PK), event_id (FK), mode, input, output, created_at

4.9.2 Entity-Relationship Diagram

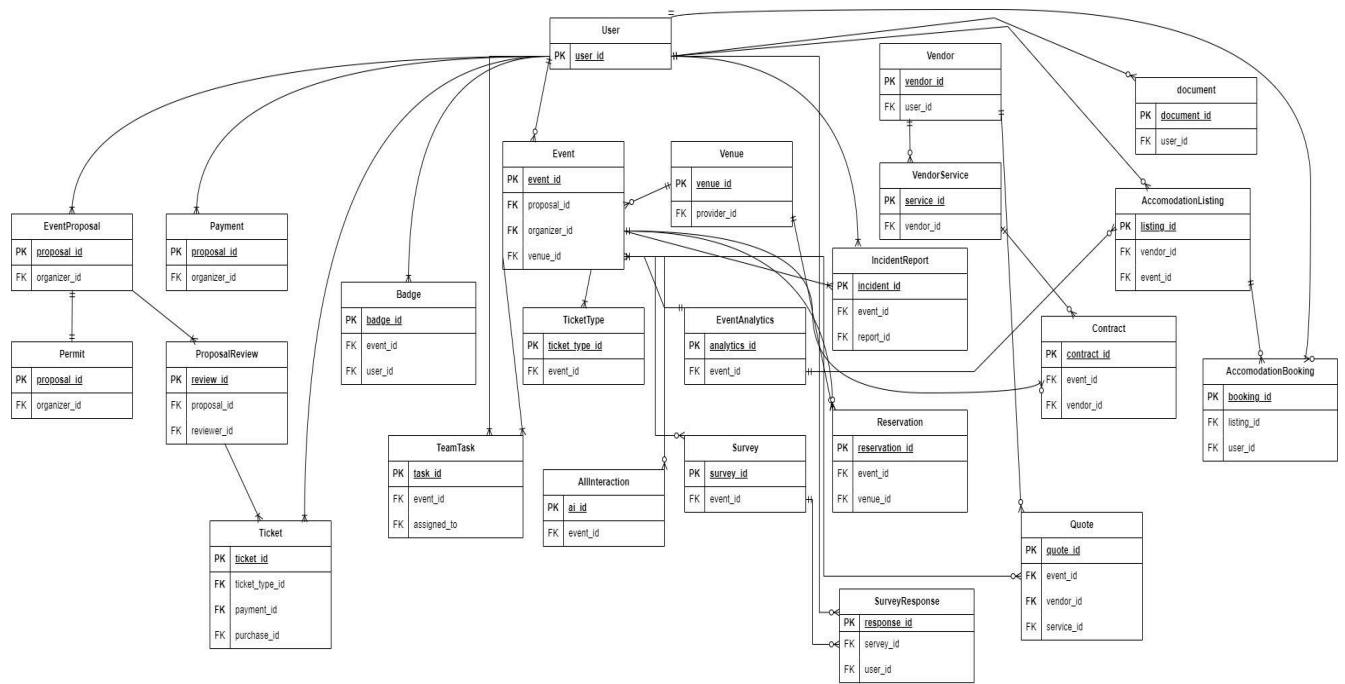


figure 37. Entity-Relationship diagram

4.10 Access Control

i. Authentication

- Single User Login Flow: All users authenticate using email and password, regardless of their role (attendee, organizer, vendor, government official, or administrator).
- Passwords are securely hashed (e.g., bcrypt) before storage to ensure data security.
- Upon successful login, a JSON Web Token (JWT) is issued for session management, providing access based on the user's role.
- For sensitive roles like government officials and administrators, multi-factor authentication (MFA) is enforced using OTPs via email or SMS.
- Session tokens expire after a set period, with refresh mechanisms for continued access.

ii. Authorization

- Role-Based Authorization: Users are granted permissions tied to their roles, aligning with their intended interactions:
- An attendee can view events, purchase tickets, and reserve accommodations.
- An organizer can submit proposals, create and publish events, generate AI schedules, and manage event details.
- A vendor can register, sign contracts, configure inventory, and enable sales for assigned events.
- A government official can review and verify proposals, approve or reject events, and handle licensing.
- An administrator has full oversight, including user management, log monitoring, and system reports.
- No dynamic adaptation beyond roles; permissions are strictly enforced via API middleware to prevent unauthorized actions.
- Contextual access for shared resources, such as event data editable only by owners or assignees.

iii. Admin Oversight

Administrator Role:

The admin is a high-privilege entity responsible for platform maintenance, user moderation, dispute resolution, and performance monitoring.

- Admins interact via a dedicated dashboard for tasks like role assignments, audit logs, and backups.
- Admin actions are fully logged to ensure accountability, separate from standard user workflows.

iv. Security Features

Self-Hosted System:

- The platform runs independently post-deployment, with no ongoing third-party dependencies, ensuring control over data and operations.
- This promotes privacy and autonomy, hosted on dedicated servers managed by the organization.

Data Access and Permissions:

- Features like proposal submissions, contract signing, or ticket purchases are limited to authenticated users in relevant roles.
- Unauthorized attempts (e.g., accessing unassigned events or modifying others' data) are blocked with error responses.

Encryption:

- All communications use HTTPS/TLS for data in transit.
- Sensitive data (e.g., personal details, contracts) is encrypted at rest using AES-256.

Future Growth Considerations:

- As the platform expands internationally, advanced features like ABAC can be added for context-based rules (e.g., location-restricted access).
- Integration with monitoring tools for real-time threat detection and automated audits.

CHAPTER 5: IMPLEMENTATION

5.1 Overview

This section provides a high-level summary of the implementation phase of the Global Connect Ethiopia project. It details the critical steps being undertaken to transform the conceptual design into a functional, intelligent event management platform. The implementation process is structured around key areas, including the adoption of rigorous coding standards, the selection of modern development tools, and the refinement of a Figma-based prototype, all of which ensure the platform's professional quality.

The development approach is divided into client-side and server-side components to ensure a scalable and modular system. On the client side, the application is built using React.js and Tailwind CSS to deliver a responsive, multilingual, and intuitive user interface. This interface empowers users to independently design, manage, and host professional events through role-specific dashboards for organizers, vendors, and government officials. Key features being implemented include a digital marketplace for vendor bookings, a centralized event coordination dashboard to prevent scheduling overlaps, and an Integrated Promotional Design Utility (IPDU) for automated marketing workflows.

On the server side, the application is developed using FastAPI (Python) and Uvicorn, facilitating the creation of a high-performance RESTful API for handling complex business logic and real-time data processing. The backend handles secure JWT-based authentication, role-based access control (RBAC), and automated workflows for event licensing and government approvals. A MongoDB database is utilized for scalable and flexible data storage, supporting everything from user profiles and event proposals to ticket inventory and financial records.

A distinctive feature of this implementation is the integration of Artificial Intelligence to support organizers. This includes the training of machine learning models to provide AI-assisted event roadmaps, automated event schedule generation, and cost estimation tools.

These intelligent services aim to revolutionize event planning in Ethiopia by providing data-driven insights and reducing the manual coordination traditionally required.

Throughout the implementation, Visual Studio Code serves as the primary integrated development environment, with Git and GitHub ensuring seamless version control and collaborative development. This phase establishes a solid foundation for a localized, enterprise-grade solution that aligns with Ethiopia's digital transformation goals.

5.2 Coding Standards

To ensure code quality, maintainability, and scalability, Global Connect Ethiopia follows standardized coding practices and established software engineering best practices throughout the implementation phase. These standards promote consistency, readability, security, and ease of future system maintenance.

General Guidelines

- The system is developed using a layered architecture with clear separation between the frontend, backend, and database layers to improve modularity and maintainability.
- Clear and meaningful naming conventions are used for variables, functions, API routes, and database fields.
- A project README is maintained to document system setup, configuration, and execution steps.
- Inline comments are added to complex or critical sections of the code for clarity.
- Standardized error-handling mechanisms are implemented to ensure system stability and user-friendly feedback without exposing sensitive information.
- Git and GitHub are used for version control, collaboration, and code review through pull requests.

Frontend Standards

The frontend is implemented using React.js with a component-based architecture to promote reusability and easier maintenance.

- Tailwind CSS : is used to maintain consistent styling and responsive layouts across devices.

- Asynchronous API communication is handled properly, with loading and error states managed to enhance user experience.

Backend Standards

Backend services are developed using FastAPI, following RESTful API design principles.

- APIs return consistent JSON responses with appropriate HTTP status codes.
- JWT-based authentication and role-based access control (RBAC) are enforced to secure system resources.
- Input validation and request handling are implemented to prevent invalid or malicious data entry.

Database Standards

- MongoDB is used for data storage, with well-structured collections reflecting system entities.
- Indexing is applied to frequently accessed fields to improve query performance.
- Application-level validation ensures data integrity and consistency.

5.3 Development Tools

The development of our system employed a variety of modern tools and technologies, organized according to their functional roles in frontend development, backend development, database management, testing, and project management. These tools support efficient development, testing, and maintenance of the system.

- Frontend Technologies

React.js - React.js used to build dynamic, modular, and responsive user interfaces. Its component-based architecture enabled reusable UI components and efficient state management throughout the system.

Tailwind CSS - Tailwind CSS will be used to provide rapid styling, consistent layout structures, and reusable UI components, ensuring a responsive and visually consistent user experience.

- Backend Technologies

FastAPI (Python) - FastAPI used to develop high-performance back-end APIs. It provides built-in data validation, automatic API documentation, and supports asynchronous request handling, making it suitable for scalable back-end services.

Uvicorn - Uvicorn served as the ASGI server for running the FastAPI backend locally. It enabled fast request handling and supported asynchronous communication between system components.

- Database Tools

MongoDB Community Edition - MongoDB used as a local NoSQL database for storing system data such as users, events, vendors, tickets, and operational records. Its flexible document-based structure supports scalable and efficient data management.

MongoDB Compass - MongoDB Compass used as a graphical user interface for managing database collections, executing queries, and monitoring database activity during development.

- Development & Design Tools

Figma - Figma used for UI/UX design activities, including wireframing, prototyping, and interface layout design prior to implementation.

Visual Studio Code (VS Code) - VS Code served as the primary integrated development environment for frontend and backend development, offering debugging tools, extensions, and version control integration.

IPDU (Integrated Promotional Design Utility) - IPDU is a custom-built tool within the system used to generate event promotional materials, supporting automated and consistent event marketing workflows.

- Testing Tools

Postman - Postman used to test backend API endpoints running on localhost, enabling verification of request handling and response correctness.

PyTest - PyTest used to perform unit and integration testing for backend components, ensuring reliability and correctness of the application logic.

Jest and React Testing Library - Jest and React Testing Library were used to test frontend components, user interface behavior, and user interactions within the React application.

Browser Developer Tools - Built-in browser developer tools were used to debug client-side scripts, inspect layouts, and analyze performance issues.

- Project Management Tools

Git and GitHub - Git and GitHub used for version control, repository hosting, collaborative development, and change tracking throughout the project lifecycle.

Trello - Trello is used to support task management, sprint planning, and Agile workflow coordination among project contributors.

5.4 Prototype

This section describes the development and refinement of the system prototype, emphasizing the progression from initial design concepts to the final prototype used on implementation.

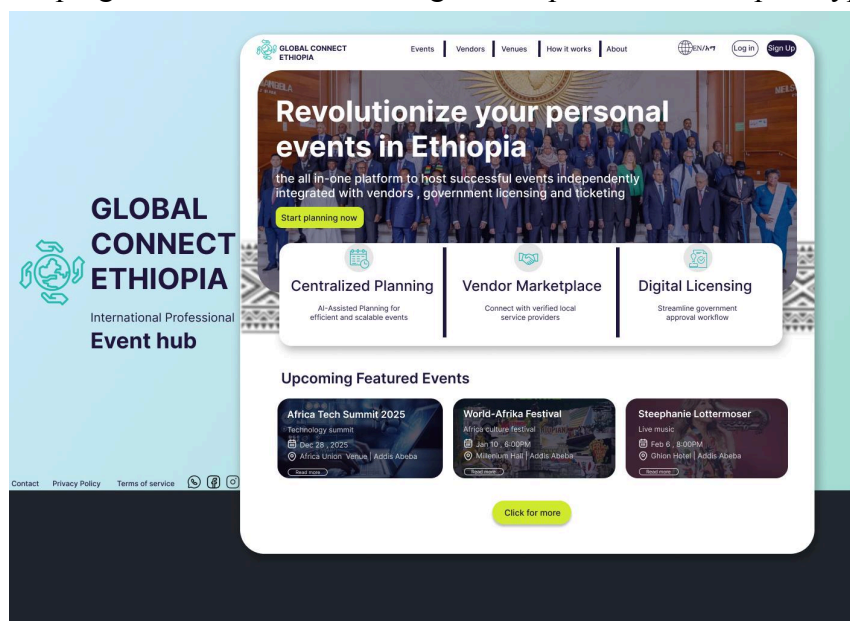


figure 38. Home page

Global Connect Ethiopia

Home Organizer Create Event Vendor Team Setting

Create New Event

1.Event info & Proposal 2.Venue & Schedule 3.Vendors & Team 4.Ticketing 5.Review & Publish

Step 1 : Basic Information and Proposal Submission

Event title Event Type

Expected Participant Budget Summary

Description

Use AI Assisted to draft description

Document Upload for Govenment Approval

Event Proposal [PDF] Personal Identification Working License

Save Draft Next: Venue & Schedule

figure 39. Event creation page

Global Connect Ethiopia

Dashboard Profile Services Requests Bookings Setting Payments

Vendor Dashboard

Requests Received 3 Active Contracts 5 Total Earnings ETB 450,000 My Rating 4.5/5.0 (50 Review)

Recent Requests

Catering for Addis Tech Summit 2025 Request from Abebe View & Quote

Ethiopia Investment Forum Request from Sara View & Quote

Upcoming Events

December 2025

December 8 - 11 Mon - Tue Cultural Expo

December 22 - 27 Mon - Tue Addis Tech Summit

Complete your profile

76% completed

figure 40. Vendor dashboard

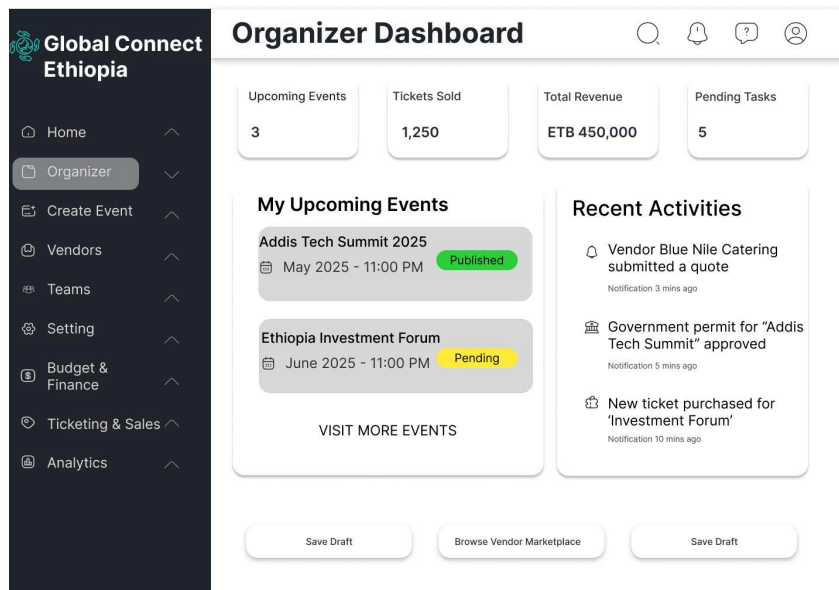


figure 41. Organizer dashboard

5.5 Implementation Detail

This section delves into the technical specifics of how the Global Connect Ethiopia platform will be built, focusing on the client-side (frontend) and server-side (backend) components.

5.5.1 Client-Side Implementation

The client-side of the platform will be developed using React.js, a JavaScript library for building user interfaces, chosen for its component-based architecture and efficient state management. This will enable a dynamic, single-page application (SPA) experience where users can interact seamlessly without full page reloads. Supporting libraries will include Redux for global state management, Tailwind CSS for pre-built responsive components, and Axios for handling HTTP requests to the backend API.

Key aspects of the client-side implementation will include:

User Interface Structure: The UI will be organized into reusable components such as dashboards, forms, and modals. The main App component will serve as the entry point, routing users based on authentication status using React Router. For example, protected routes (e.g., event creation) will require JWT tokens stored in localStorage.

Authentication and Registration: Users will register or log in via forms that validate inputs (e.g., email format, password strength) using libraries like `crypto-bcrypt`. Upon successful authentication, the backend will return a JWT, which will be used for subsequent API calls. This component will handle form submission, API interaction, and error display, ensuring a user-friendly experience.

Event Management Features: Organizers will be able to submit event proposals through a multi-step form that collects details like title, date, venue, and description. Real-time validation will prevent errors, and upon submission, data will be sent to the backend for ministry review. The dashboard will display event status with interactive elements like progress bars.

AI Integration on Client-Side: For features like AI-generated schedules, the client will send user inputs (e.g., event type, duration) to a backend endpoint. The response will be rendered dynamically, e.g., as a timeline using libraries like `React Calendar` or custom SVG components.

Vendor and Ticketing Interfaces: Vendors will register via a dedicated form, uploading documents for verification. Attendees will browse events and purchase tickets using integrated payment gateways (e.g., Stripe or local Ethiopian services like Chapa). The UI will update inventory in real-time via WebSockets for live availability.

Responsiveness and Accessibility: Media queries will ensure mobile compatibility, and ARIA attributes will improve accessibility for screen readers. Testing will be conducted with tools like Lighthouse to achieve high performance scores (e.g., >90% for accessibility).

The client-side code will be bundled using Webpack for production, minimizing load times through code splitting and lazy loading.

5.5.2 Server-Side Implementation

The server-side will be implemented with FastAPI, a Python-based framework for building APIs, selected for its high performance, automatic Swagger documentation, and async support. It will handle business logic, data validation, and integrations with the MongoDB database and external services. Security features will include JWT authentication, rate limiting, and input sanitization to prevent common vulnerabilities like SQL injection (though NoSQL mitigates this).

Key aspects will include:

API Structure: The backend will expose RESTful endpoints organized by modules (e.g., /api/auth, /api/events, /api/vendors). FastAPI's dependency injection will ensure authenticated routes require valid JWTs.

Authentication Endpoint: It will handle user registration and login, hashing passwords with bcrypt and generating JWTs. This will secure user sessions and integrate with the client-side.

Event Proposal and Review: Endpoints like /api/events/submit will validate proposals using Pydantic models and store them in MongoDB with a 'pending' status. A simulated ministry review process will trigger email notifications via SMTP or integrated services like SendGrid.

AI-Assisted Features: For schedule generation, the server will integrate with AI libraries (e.g., OpenAI API or custom ML models via scikit-learn). An endpoint will receive event details, process them, and return optimized schedules.

Database Interactions: Using PyMongo, the server will perform CRUD operations. For example, events will be stored in a collection with fields like _id, title, status, and schedule. Aggregation pipelines will handle complex queries, such as analytics on ticket sales.

Integrations and Security: Payment processing will integrate with APIs Stripe, while file uploads (e.g., vendor documents) will use secure storage (e.g., AWS S3). CORS will be configured for client-server communication, and logging with structlog will capture errors for debugging.

Conclusion

In conclusion, the Global Connect Ethiopia project successfully addresses critical challenges within Ethiopia's event management ecosystem by introducing a unified, client-driven, and digitally integrated platform. The system overcomes the limitations of fragmented coordination, manual licensing processes, unregulated ticketing, and high organizational costs that characterize existing solutions. By consolidating event creation, vendor management, ticketing, accommodation, government approvals, and post-event analytics into a single web-based platform, Global Connect Ethiopia enhances transparency, efficiency, and control throughout the entire event lifecycle.

The project also demonstrates the effective application of modern software engineering principles and technologies. Through the use of Agile methodology, React, FastAPI, and MongoDB, the system achieves modularity, scalability, and usability while maintaining security and performance standards suitable for professional and high-profile events. Features such as AI-assisted scheduling, role-based dashboards, digital licensing workflows, and analytics tools highlight the system's innovation and practical relevance. From an academic perspective, the project serves as a comprehensive capstone that integrates theoretical knowledge with real-world problem solving, system design, and full-stack implementation.

Overall, Global Connect Ethiopia contributes significant technological, socio-economic, and institutional value by supporting local vendors, empowering organizers, and aligning with Ethiopia's national digital transformation goals. Although the current implementation is limited to a web-based prototype and urban contexts, it establishes a strong foundation for future enhancements, including mobile applications, deeper government integration, and large-scale deployment. With further development and adoption, Global Connect Ethiopia has the potential to position Ethiopia as a regional hub for professional events and international collaboration while fostering sustainable growth in the local digital economy.

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